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Grade	**0.00** out of 150.00 (**0**%)

Question 1

Not answered

Marked out of 5.00

Theme- Functional Brain Networks II

Lead In: For each of the following regions, choose the specified functional brain network from the above list to which they belong as a functional unit.

1. Precuneus ✖
2. Posterior Cingulate Cortex ✖
3. Anterior Insula ✖
4. Intraparietal sulcus ✖
5. Dorsal Anterior Cingulate Cortex ✖

Auditory Network
Central Executive Network
Default Mode Network
Mesolimbic Network
Mirror Neuron System
Papez Circuit
Salience Network
Sensorimotor Network
Tuberoinfundibular Network
Visual Network

Explanation:

The default mode network includes the posterior cingulate cortex (PCC), precuneus, medial prefrontal cortex (mPFC) and angular gyrus. These predominantly midline cortical structures play a major role in self-processing, internally oriented thinking and processing information not relevant to a task at hand.

The dorsal attention network, also called the frontoparietal or central executive network, comprises of dorsolateral prefrontal cortex, intraparietal sulcus (IPS) as well as lateral superior parietal lobule (together called posterior parietal cortex – PPC) and middle frontal gyrus/premotor junction (equivalent of the frontal eye field in primates). These predominantly lateral cortical structures play a major role in executive processing, externally oriented thinking and processing information relevant to a task at hand.

The salience network comprises of the anterior prefrontal cortex (aPFC), dorsal anterior cingulate cortex (dACC), anterior insula and frontal operculum, with some subcortical structures (thalamus and putamen).

See

for a short illustrative description.

The correct answer is:

1. Precuneus - Default Mode Network
2. Posterior Cingulate Cortex - Default Mode Network
3. Anterior Insula - Salience Network
4. Intraparietal sulcus - Central Executive Network
5. Dorsal Anterior Cingulate Cortex - Salience Network

Question 2

Not answered

Marked out of 5.00

Theme- Perturb and measure techniques II
Lead In: For each of the techniques below choose the most appropriate cellular function or property measured by the technique.

1. Extracellular recording (Choose TWO) ✖
2. Intracellular recording (Choose TWO) ✖
3. Patch-clamp technique ✖

Exictatory Postsynaptic Potential
Gene silencing
Histone modification
Inhibitory Postsynaptic Potential
Ion channel function
Local Field Potential
Neuronal spiking
Myelin quantity

Explanation:

Electrophysiology techniques are often categorised by where the recording instrument, the electrode, is placed in the neural specimen. In an **extracellular recording** experiment, an electrode is placed just outside a neuron of interest. The goal of many electrophysiology experiments is to quantify the frequency of action potentials, or **spikes**, that neuron fires in response to a specific stimulus. Usually, these experiments use extracellular recording electrodes. Extracellular recordings cannot report detailed information about subthreshold potential changes such as excitatory postsynaptic potentials (EPSPs) or inhibitory postsynaptic potential (IPSPs). Instead, extracellular recordings can detect **local field potentials,** the sum of postsynaptic activity of multiple neurons within a volume of neural tissue, to determine activity changes in a population of neurons.

In an **intracellular recording** experiment, an electrode is inserted inside a neuron of interest. This can be used to measure subthreshold changes potential changes, such as **EPSPs or IPSPs.**

Finally, using **patch clamp techniques**, an electrode is closely opposed to a neuronal membrane, forming a tight seal with a patch of the membrane. Questions regarding the 'open' and 'closed' states of a **specific ion channel** are best addressed using patch clamp techniques.

Ref: Carter M, Essner R, Goldstein N, Iyer M. Guide to Research Techniques in Neuroscience, 3rd Edition. 2022. pp. 85-102.

The correct answer is:

1. Extracellular recording (Choose TWO) - Local Fields Potentials, Neuronal spiking
2. Intracellular recording (Choose TWO) - Excitatory Postsynaptic Potential, Inhibitory Postsynaptic Potential
3. Patch-clamp technique - Specific ion channel function

Question 3

Not answered

Marked out of 6.00

Theme- Animal Models of Psychiatric Disorders I

Lead In: For each of the following conditions, choose the appropriate animal model from the above list

1. Schizophrenia (Choose FOUR)

2. OCD (Choose TWO)

- Aromatase Knockout
- BDNF haploinsufficient mice
- Chronic restraint
- Circadian rhythm distruption
- Dysbindin knockout
- ERK1 knockout
- Frontal ablation
- Maternal immnue Activation
- Maternal Separation
- Neonatal clomipramine
- Neonatal ventral hippocampal lesion
- Olfactory bulbectomy
- Social Defeat
- Subacute ketamine administration

Explanation:

Maternal immune activation (MIA) is induced in pregnant rodents with agents like the viral mimetic polyinosinic-polycytidylic acid (poly I: C). A rather large body of evidence indicates that the offspring of rodents and primates that have undergone MIA during pregnancy show anatomical, neurochemical, electrophysiological, and behavioural changes consistent with schizophrenia, as well as other neurodevelopmental disorders, including autism.

Another developmental preparation used to study schizophrenia involves neonatal lesions of the rat ventral hippocampus on postnatal day 7 via local injection of either the excitotoxin ibotenic acid or temporary inactivation via tetrodotoxin. Neonatal lesions, but not similar lesions conducted in early adulthood, lead to an array of abnormalities consistent with schizophrenia that emerge at various times after puberty, similar to the typical onset of schizophrenia in late adolescence/early adulthood.

Dysbindin is a synaptic protein implicated in the regulation of exocytosis, vesicle biogenesis, and receptor trafficking processes involved in excitatory synaptic transmission. A reduction in dysbindin expression affects NMDA receptor function and impairs working memory. Dysbindin mutant mice show behaviours consistent with a schizophrenia-like phenotype, including hyperactivity, learning and memory deficits, and increased impulsive and compulsive behaviours.

The drugs most frequently used in animal models of schizophrenia are dopamine enhancers (e.g., amphetamine and apomorphine) and non-competitive NMDA receptor antagonists (PCP, ketamine, and MK-801).

Aromatase knockout mice were originally developed to study the role of estradiol in the sexual differentiation of the reproductive system. They lack a functioning aromatase enzyme and are therefore estrogen-deficient. Male, but not female, knockout mice exhibit increased wheel-running activity and grooming but decreased ambulatory activity.

Andersen et al developed a multiple OCD-like behaviour model in rats. They compared rats treated with 16 intraperitoneal injections of clomipramine administered across postnatal days 9-16 with those receiving a saline vehicle. Clomipramine-exposed rats showed more anxiety-like behaviour in the elevated plus maze and a directed anxiety response in the marble-burying task, burying more foreign/novel objects. They also showed more perseveration in a reversal task, a general impairment of discrimination learning, and increased hoarding behaviour.

Ref: Winship IR et al. An overview of animal models related to schizophrenia. The Canadian Journal of Psychiatry. 2019 Jan;64(1):5-17.

Aguilar-Valles A et al. Maternal immune activation and the development of dopaminergic neurotransmission of the offspring: relevance for schizophrenia and other psychoses. Frontiers in Psychiatry. 2020 Aug 21;11:852.

Alonso P et al. Animal models of obsessive–compulsive disorder: utility and limitations. Neuropsychiatric disease and treatment. 2015;11:1939.

The correct answer is:

1. Schizophrenia (Choose FOUR) - Maternal immune activation, Neonatal ventral hippocampal lesion, Dysbindin knockout, Subacute ketamine administration

2. OCD (Choose TWO) - Aromatase knockout, Neonatal clomipramine

Question 4

Not answered

Marked out of 3.00

Theme: Class of medications
Lead In: Which of the following classes of antipsychotics are each of the below medications within?

1. Haloperidol ✖
2. Sulpiride ✖
3. Olanzapine ✖

Butyrophenones
Diphenylbutylpiperidines
Phenothiazines
Thioxanthenes
Benzamides
Tricyclics
Benzisoxazoles
Phenylpiperazines

Explanation:

Haloperidol is an example of a butyrophenone and is the only one now used widely. Sulpiride and amisulpride are examples of benzamides. Olanzapine is a tricyclic antipsychotic, as are clozapine, quetiapine, loxapine zotepine and asenapine. Note that olanzapine is a thienobenzodiazepine but it originated from the tricyclic structure of clozapine by incorporating a thiophene as a replacement for one of the benzo rings.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 3.

The correct answer is:

1. Haloperidol - Butyrophenones

2. Sulpiride - Benzamides

3. Olanzapine - Tricyclics

Question 5

Not answered

Marked out of 3.00

Akathisia
Parkinsonism
Tardive dyskinesia
Serotonin syndrome
Neuroleptic malignant syndrome
Anticholinergic effects

Theme- Medication side-effects
Lead In: Which of the above medication side-effects are described in each of the following scenarios?

1. Mrs. Freda Charleston is a 57-year-old woman with a diagnosis of paranoid schizophrenia who has been taking antipsychotics for many years and is now experiencing involuntary movements of her tongue and lips. ✖

2. Mr. Henry Adams is a 28-year-old man with a diagnosis of first episode psychosis who has been started on an antipsychotic medication. He is now describing feeling like his limbs are stiff, that it takes him longer to get out of bed and that he is drowsy during the day. ✖

3. Mr Richard Stevens is a 23-year-old man who describes feeling agitated, cannot stand still, and feels like he is going to swallow his tongue. ✖

Explanation:

In 80% of cases of **tardive dyskinesia** (TD), especially at the milder end, disorder is found in the lower third of the face. It has always been thought of as starting with the tongue. Early signs are usually described as vermicular (worm-like) and in the tongue. In the early stages individual parts of the tongue's surface can be seen to move, with progression, turning it into a 'bag of worms'. Mouth-opening, forced jaw closure, bruxism, lateral jaw deviation and a range of lip movements (including puckering, pouting, smacking, and bridling), with or without accompanying sounds, complete a characteristic picture.

When D2 receptors are blocked acutely in the nigrostriatal pathway – the same pathway that degenerates in Parkinson disease – this can cause a condition known as drug-induced **parkinsonism** because it looks similar to Parkinson disease with tremor, muscle rigidity, and slowing of movements (bradykinesia) and loss of movement (akinesia).

Akathisia is a syndrome of motor restlessness seen commonly in treatment with D2 blockers. Akathisia has both subjective and objective features. Subjectively, there is a sense of inner restlessness, mental unease, or dysphoria. Objectively, there are restless movements, most typically being lower-limb movements such as rocking from foot to foot, walking or marching in place when standing, or pacing.

Ref: Owens DC. Tardive dyskinesia update: the syndrome. BJPsych Advances. 2019 Jan;25(1):57-69.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 165, 169.

The correct answer is:

1. Mrs. Freda Charleston - Tardive dyskinesia
2. Mr. Henry Adams - Parkinsonism
3. Mr. Richard Stevens - Akathisia

Question 6

Not answered

Marked out of 4.00

Theme- Equivalent benzodiazepine doses

Lead In: 10mg of Diazepam is equivalent to which dose of:

1. Lorazepam ✖
2. Chlordiazepoxide ✖
3. Nitrazepam ✖
4. Temazepam ✖

0.5 mg
1 mg
2 mg
5 mg
10 mg
20 mg
25 mg
40 mg
45mg

Explanation:

The approximate equivalent doses for Diazepam **5mg** are:
Alprazolam 250 micrograms
Clobazam 10mg
Clonazepam 250 micrograms
Flurazepam 7.5-15 mg
Chlordiazepoxide 12.5 mg
Loprazolam 0.5-1 mg
Lorazepam 500 micrograms
Lormetazepam 0.5-1 mg
Nitrazepam 5mg
Oxazepam 10 mg
Temazepam 10 mg

Ref: British National Formulary. Hypnotics and anxiolytics. Accessed online via https://bnf.nice.org.uk/treatment-summary/hypnotics-and-anxiolytics.html on 24 April 2020.

The correct answer is:

1. Lorazepam - 1mg
2. Chlordiazepoxide - 25mg
3. Nitrazepam - 10mg
4. Temazepam - 20mg

Lorazepam
Sertraline
Mirtazapine
Aripiprazole
Valproate
Diazepam
Carbamazepine
Quetiapine
Zolpidem
Bupropion

Question 7

Not answered

Marked out of 3.00

Theme- Side-effects
Lead In: Which of the following descriptions match the side-effects of the above medications?

1. Mrs Paula Taylor is a 42-year-old woman with a diagnosis of paranoid schizophrenia. She has recently changed her medication and her husband now reports that she often complains of an unquiet feeling in her legs and continuous movement of her legs preventing her from falling asleep. ✖

2. Miss Lauren Atkins is a 23-year-old woman who has recently been started on a new medication. She has now attended A&E after developing marked restlessness, sweating and some confusion. She is noted to have ankle clonus on examination. ✖

3. Mr Simon Hughes is a 70 old man who has attended his GP after taking medication for poor sleep but now reports daytime drowsiness and taking more time than usual to get up in the morning due to sedation. ✖

Explanation:

1) Among antipsychotics, olanzapine and quetiapine have been the most frequently reported agents associated with restless leg syndrome (RLS) symptoms. The mechanism of antipsychotic-induced RLS is suspected to be dopaminergic dysfunction. This may be attributed to the medications' variable affinity for the D2 receptor as well as serotonin receptor activity.

2) Serotonin syndrome, a potentially life-threatening reaction to serotonergic drugs, is most frequently observed in cases of antidepressant overdose and in cases where individuals are prescribed multiple medications with serotonergic actions. MAOIs have been associated with the condition, particularly in cases where they are used in combination with other serotonergic medication. However, cases have also been observed after a single dose of an SSRI.

3) Diazepam generally has an elimination half-life of 20-50 hours. Older adults have more body fat, less body water, and less albumin than

younger adults. This leads to an increased volume of distribution and a longer duration of action for some fat-soluble drugs (e.g. diazepam), higher concentrations of drugs at the site of action (e.g. digoxin), and a reduction in the amount of drug bound to albumin (increased amounts of active 'free drug'; e.g. warfarin, phenytoin) in the elderly.

Ref: Patatanian E, Claborn MK. Drug-induced restless legs syndrome. Annals of Pharmacotherapy. 2018 Jul;52(7):662-72.

Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. p. 683.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 231.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 602.

The correct answer is:

1. Mrs Paula Taylor - Quetiapine
2. Miss Lauren Atkins - Sertraline
3. Mr Simon Hughes - Diazepam

Question 8

Not answered

Marked out of 5.00

Theme- Mechanism of action

Lead In: Which of the following best describes the mode of action for each of the below medications?

1. Propranolol ✖
2. Buproprion ✖
3. Diazepam ✖
4. Zopiclone ✖
5. Baclofen ✖

Noradrenaline and dopamine reuptake inhibitor.
Beta-1-adrenaline antagonist
GABA-A agonist
Beta-1 and Beta-2 adrenergic antagonist.
GABA-A and B-agonist.
Serotonin and dopamine reuptake inhibitor.

Explanation:

Propranolol is a non-selective beta-adrenergic antagonist with most action at the beta-1 and -2 receptors. Bupropion is a dopamine and noradrenaline reuptake inhibitor. Benzodiazepines and the Z-drugs act as agonists at the GABA-A and thus enhance the actions of GABA. Baclofen is a GABA analogue, which acts as a selective agonist of GABA-B receptors, and is used as a muscle relaxant.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 229, 653, 879.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 303.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 599.

The correct answer is:

1. Propranolol - Beta-1- and beta-2-adrenergic antagonist

2. Bupropion - Noradrenaline and dopamine reuptake inhibitor
3. Diazepam - GABA-A agonist
4. Zopiclone - GABA-A agonist
5. Baclofen - GABA-B agonist

Question 9

Not answered

Marked out of 4.00

Theme: Hormones and areas of secretion
Lead In: Which of the following areas secrete each of the below hormones?

1. Cortisol ✖
2. Leptin ✖
3. Prolactin ✖
4. Somatostatin ✖

Pituitary
Adrenal gland
Adipose Tissue
Hypothalamus
Anterior pituitary
Posterior pituitary

Explanation:

Cortisol is produced mainly by the zona fasciculata of the adrenal cortex within the adrenal gland. Leptin is predominantly made by adipose cells and enterocytes in the small intestine. Prolactin is produced mainly in the anterior pituitary but also in the uterine decidua, myometrium, breast, lymphocytes, leukocytes and prostate. Somatostatin, also known as growth hormone-inhibiting hormone (GHIH) is produced by neuroendocrine neurons of the ventromedial nucleus of the hypothalamus and also by delta cells at several locations in the digestive system, namely the pyloric antrum, the duodenum and the pancreatic islets.

The correct answer is:

1. Cortisol - Adrenal gland
2. Leptin - Adipose tissue
3. Prolactin - Anterior pituitary
4. Somatostatin - Hypothalamus

Question 10

Not answered

Marked out of 4.00

Narcissistic personality disorder
Paranoid personality disorder
Borderline personality disorder
Schizotypal personality disorder
Antisocial personality disorder
Avoidant personality disorder
Schizoid personality disorder
Dependent personality disorder

Theme- Personality disorders:
Lead In: Which of the above match the below descriptions?

1. A person who is socially detached and has a lack of interest in others?

2. A person who can have short-lived psychotic episodes and experience dissociative symptoms while under stress?

3. A person who has inflated self-importance and need for admiration?

4. A person who has a callous lack of concern for others and will not pay debts?

Explanation:

1) People with schizoid personality disorder are emotionally cold, detached, and have a lack of interest in others.
2) People with borderline personality disorder can have short-lived psychotic episodes (rather than full-blown psychotic breaks),and the symptoms are circumscribed, fleeting, or questionable. Paranoia and dissociation is usually due to stress.
3) Narcissistic personality disorder is characterised by grandiosity, lack of empathy, and need for admiration.
4) A person with antisocial personality disorder shows a callous lack of concern for others, irresponsibility, irritability, aggression, inability to maintain enduring relationships, and disregard and violation of others' rights.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 523.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 564.

The correct answer is:

1. Socially detached and lack of interest - Schizoid personality disorder

2. Short-lived psychotic episodes and dissociative symptoms - Borderline personality disorder
3. Inflated self-importance and need for admiration - Narcissistic personality disorder
4. Callous lack of concern for others - Antisocial personality disorder

Question 11

Not answered

Marked out of 2.00

1 week
2 weeks
1 month
3 months
6 months
12 months
24 months

Theme- Duration of disorder

Lead In: Which of the choices best describes the duration of symptoms necessary for the diagnosis of the following disorders in ICD-11?

1. Miss Kelly Pearson is a 23-year-old woman who attends her GP with complaints of persistently unstable mood. At times her mood is elevated and she has lots of energy, but at other times she feels down and uninterested in doing anything. Though there are brief times when she feels fine, these symptom-free periods are not prolonged. ✖

2. Mr Robert Seymour is a 45-year-old man who has been attending his GP for some time with complaints of feeling 'kind of down'. He isn't particularly interested in doing much of anything and generally has a low view of himself. Every once in awhile he feels better for a week or two, but then the low feelings return. ✖

Explanation:

1) In ICD-11, cyclothymic disorder is characterised by a persistent instability of mood over a period of least **2 years**, involving numerous periods of hypomanic (e.g., euphoria, irritability, or expansiveness, psychomotor activation) and depressive (e.g., feeling down, diminished interest in activities, fatigue) symptoms that are present during more of the time than not. While brief symptom-free intervals are consistent with the diagnosis, there have never been any prolonged symptom-free periods (e.g., lasting 2 months or more) since the onset of the disorder.

2) In ICD-11, dysthymic disorder is characterised by a persistent depressive mood (i.e. lasting **2 years** or more), for most of the day, for more days than not, as reported by the individual (e.g., feeling down, sad) or as observed (e.g., tearful, defeated appearance). The depressed mood is accompanied by additional symptoms such as markedly diminished interest or pleasure in activities, reduced concentration and attention or indecisiveness, low self-worth or excessive or inappropriate guilt, hopelessness about the future, disturbed sleep or increased sleep, diminished or increased appetite, or low energy or fatigue. While brief symptom-free intervals during the period of persistent depressed mood are consistent with the diagnosis, there have never been any prolonged symptom-free periods (e.g., lasting 2 months or more) since the onset of the disorder.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed, ICD-11; World Health Organization, 2020)

The correct answer is:

1. Miss Kelly Pearson - 24 months

2. Mr Robert Seymour - 24 months

Question 12

Not answered

Marked out of 1.00

In comparison to DSM-5, ICD-11 has the following features, except:

Select one:

- Schizotypal disorder is included under 'schizophrenia or other primary psychotic disorders'
- Symptom qualifiers for schizophrenia include the presence of positive, negative, depressive, manic, psychomotor, and cognitive symptoms
- 'Acute and transient psychotic disorder' is a recognised diagnosis
- A dimensional approach to the classification of personality disorders has been adopted
- 6 months duration of symptoms is required for the diagnosis of schizophrenia

Your answer is incorrect.

Explanation:

All the statements are true of ICD-11 except for the duration criteria for schizophrenia. In ICD-11, symptoms must have persisted for at least one month in order for a diagnosis of schizophrenia to be assigned.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed, ICD-11; World Health Organization, 2020)

Bach B, First MB. Application of the ICD-11 classification of personality disorders. BMC psychiatry. 2018 Dec;18(1):1-4.

Stein DJ et al. Mental, behavioral and neurodevelopmental disorders in the ICD-11: an international perspective on key changes and controversies. BMC medicine. 2020 Dec;18(1):1-24.

The correct answer is: 6 months duration of symptoms is required for the diagnosis of schizophrenia

Question 13

Not answered

Marked out of 1.00

Which of the following features validate a diagnosis of OCD when a patient presents with repetitive thoughts?

Select one:

- Thoughts spoken aloud
- Resisting the thoughts
- Possessing full control of the thoughts
- Thoughts coming from an external agent
- Repeating thoughts take a variety of new themes each day

Your answer is incorrect.

Explanation:

Obsessive-compulsive disorder is characterised by the presence of persistent obsessions or compulsions, or most commonly both. Obsessions are repetitive and persistent thoughts, images, or impulses/urges that are intrusive, unwanted, and are commonly associated with anxiety. The return of the obsession can be resisted for a time, at the expense of mounting anxiety. The individual attempts to ignore or suppress obsessions or to neutralise them by performing compulsions.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 115.

The correct answer is: Resisting the thoughts

Question 14

Not answered

Marked out of 1.00

Which of the following distinguishes depressive reaction of adjustment disorder from major depressive disorder?

Select one:

- Resolution is seen within 6 months after termination of the stressor
- Symptoms develop within 1 month of the onset of the stressor
- Marked psychosocial and occupational impairment in functioning is seen
- Symptoms are distinctly different from those seen in depressive episode
- Follows a stressful life event

Your answer is incorrect.

Explanation:

Adjustment disorder is characterised by a maladaptive reaction to identifiable psychosocial stressors, and can include depressive symptoms (e.g., rumination). The symptoms of an adjustment disorder typically resolve within 6 months unless the stressor persists for a longer duration. The majority of depressive episodes show remission within 1 year.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

The correct answer is: Resolution is seen within 6 months after termination of the stressor

Question 15

Not answered

Marked out of 1.00

A patient complains of poor sleep due to uncomfortable sensations in her legs at night. The most probable diagnosis is

Select one:

- Akathisia
- Anxiety
- Periodic limb movement disorder
- Dyskinesia
- Restless legs syndrome

Your answer is incorrect.

Explanation:

The International Classification of Sleep Disorders (ICSD-3) defines restless legs syndrome (RLS) as an urge to move the legs, usually accompanied or thought to be caused by uncomfortable and unpleasant sensations in the legs that: (1) Begin or worsen during periods of rest or inactivity: (2) Are partially or totally relieved by movement; (3) Occur exclusively or predominantly in the evening or night, rather than during the day. RLS symptoms cause concern, distress, sleep disturbance, or other functional impairment.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 474.

The correct answer is: Restless legs syndrome

Question 16

Not answered

Marked out of 1.00

Who first described Bulimia Nervosa?

Select one:

- Klermann and Weissmann
- Greisinger
- Russell
- Liddle
- Crow

Your answer is incorrect.

Explanation:

Bulimia nervosa was first described in 1979 by British psychiatrist Gerald Russell as a ‘chronic phase of anorexia nervosa’ in which patients overeat and then use compensatory mechanisms such as self-induced vomiting, laxatives, or prolonged periods of starvation. 'Russell sign' is a classical skin marker of induced vomiting.

Ref: Castillo M, Weiselberg E. Bulimia nervosa/purging disorder. Current problems in pediatric and adolescent health care. 2017 Apr 1;47(4):85-94.

The correct answer is: Russell

Question 17

Not answered

Marked out of 1.00

Which of the following personality disorders was not included in either ICD-10 or DSM-5?

Select one:

- Borderline
- Paranoid
- Histrionic
- Sadistic
- Schizoid

Your answer is incorrect.

Explanation:

Sadism, a term introduced by Krafft-Ebing late in the 19th century, originally referred to sexual pleasure derived through inflicting pain and suffering on others. Over time, the term was expanded to include nonsexual enjoyment derived from sadistic acts. Sadists are believed to be motivated to a significant degree by the pursuit of pleasure, control, or satisfaction. From an interpersonal standpoint, sadistic individuals are noted to be harsh, hostile, manipulative, lacking in empathy, cold-hearted, and abrasive to those they deem to be their inferiors. Their cognitive nature is considered rigid and prone to social intolerance, and they are fascinated by weapons, and infamous crimes or perpetrators of atrocities.

Sadistic personality disorder (SPD) was included in the appendix of DSM-III-R under 'Proposed Categories Requiring Further Study', but not included in DSM-IV. It was not included in ICD-10 and is not present in ICD-11 or DSM-5. The most common criticisms of SPD as a diagnosis are the degree of overlap with other personality disorders, the paucity of data confirming its reliability and validity, and a perceived potential

for misuse (e.g., to lessen criminal responsibility). SPD is difficult to study, as such individuals tend to live in relative anonymity, do not commonly seek treatment for their sadistic traits, and conceal their pathology when the social and political climate is not conductive to such behaviour.

Ref: Myers WC et al. Sadistic personality disorder and comorbid mental illness in adolescent psychiatric inpatients. Journal of the American Academy of Psychiatry and the Law Online. 2006 Jan 1;34(1):61-71.

The correct answer is: Sadistic

Question 18

Not answered

Marked out of 1.00

Mr Wells has spent most of his life researching about a time machine. He is unmarried and has poor communication skills. He has extensive theories about extraterrestrial species. He admits to unusual experiences wherein he is able to communicate with some extraterrestrial species using clairvoyance. He is most likely to have

Select one:

- Paranoid schizophrenia
- Schizoid personality disorder
- Schizotypal personality disorder
- Cyclothymic personality
- Paranoid personality disorder

Your answer is incorrect.

Explanation:

'Schizotypal disorder' is classified along with schizophrenia and related disorders in ICD-11, but 'schizotypal personality disorder' is included with cluster A/'odd-eccentric' personality disorders in DSM-5. It shares some of the same features of schizophrenia, but not the delusions or hallucinations. Symptoms include ideas of reference, excessive social anxiety, odd beliefs or magical thinking, unusual perceptions (e.g. illusions), odd/eccentric behaviour or appearance, no close friends/confidants, odd speech, inappropriate or constricted affect, and suspiciousness.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 228-229.

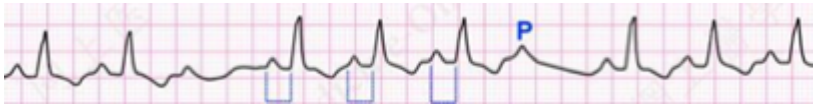
The correct answer is: Schizotypal personality disorder

Question 19

Not answered

Marked out of 1.00

Read the strip of ECG shown below and identify the relevant abnormality. Blue lines indicate PR intervals. A P wave is also marked in the figure.



Select one:

- Type I second degree heart block
- Normal ECG
- Mobitz Type II
- Complete heart block
- Mobitz type I

Your answer is incorrect.

Mobitz type 2 block: If the PR interval is fixed and normal but occasionally a P wave fails to produce a QRS complex, the patient has Mobitz type 2 block. (Ref: Houghton & Gray, Making sense of the ECG. Pg 120-122)

The correct answer is: Mobitz Type II

Question 20

Not answered

Marked out of 1.00

Choose one advantage that a MRI scan has over a CT scan of the brain:

Select one:

- MRI scan is cheaper to obtain than CT scans
- MRI scan can get multiple views of brain anatomy
- MRI scan is very useful for calcified lesions
- MRI scan can be done in emergencies
- MRI scan produces less claustrophobia

Your answer is incorrect.

Explanation:

With the introduction of MRI scanning, CT scans have been supplanted as the nonemergency neuroimaging study of choice. The only component of the brain better seen on CT scanning is calcification, which may be invisible on MRI. MRI images can be in the axial, coronal, or sagittal planes. MRI involves enclosing a patient in a narrow tube, in which the patient must remain motionless for up to 20 minutes. A significant number of patients cannot tolerate the claustrophobic conditions of routine MRI scanners and may need an open MRI scanner, which has less power and thus produces images of lower resolution.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 70-71.

The correct answer is: MRI scan can get multiple views of brain anatomy

Question 21

Not answered

Marked out of 1.00

A man was admitted with severe depression and on examination has bilateral ptosis and reduced muscle tone. Which of the following medical condition is most likely?

Select one:

- Myasthenia gravis
- Addison's disease
- Carcinoid tumors
- Diabetes
- Hypothyroidism

Your answer is incorrect.

Explanation:

Myasthenia gravis (MG) is an autoimmune disease that affects the postsynaptic membrane at the neuromuscular junction. The predominant manifestation is muscle weakness. Extraocular muscles are frequently affected, usually bilaterally but asymmetrically, with typical symptoms being intermittent drooping of the upper eyelid (ptosis) and double vision (diplopia). Patients are at least somewhat 'frustrated', as well as sometimes overwhelmed or depressed.

Ref: Gilhus NE et al. Myasthenia gravis (Primer). Nature Reviews: Disease Primers. 2019;5(1).

The correct answer is: Myasthenia gravis

Question 22

Not answered

Marked out of 1.00

Mr. Y is a 73-year-old man who initially presented with difficulty speaking and constructing sentences, particularly struggling with pronouncing longer words. In time, he has also developed parkinsonism, difficulty swallowing, and apathy. What is the most likely diagnosis?

Select one:

- Behavioural variant frontotemporal dementia (bvFTD)
- Progressive supranuclear palsy (PSP)
- Semantic variant primary progressive aphasia (svPPA)
- Corticobasal degeneration syndrome (CBS)
- Nonfluent variant primary progressive aphasia (nfvPPA)

Your answer is incorrect.

Explanation:

The frontotemporal dementia (FTD) spectrum encompasses three variant syndromes, classified according to their leading features. Behavioural variant FTD (bvFTD) presents with progressive decline in emotional reactivity and interpersonal and executive skills, associated with a variable profile of frontotemporal atrophy. Semantic variant primary progressive aphasia (svPPA) presents with loss of vocabulary and ultimately degrades semantic knowledge of sensory objects and nonverbal concepts as well as words. Nonfluent primary progressive aphasia (nfvPPA) presents with impaired motor speech production and/or sentence construction. This syndrome (nfvPPA) is more likely than other FTD syndromes to present in later life. There is often a history of mispronouncing longer words and sometimes reappearance of a childhood stutter. A substantial proportion of nfvPPA patients develop parkinsonism and features of the CBS-PSP spectrum and swallowing becomes affected. Executive dysfunction, apathy, and depression are also frequent issues.

Ref: Sivasathiaseelan et al., Frontotemporal dementia: a clinical review. In Seminars in neurology 2019 Apr (Vol. 39, No. 02, pp. 251-263). Thieme Medical Publishers.

The correct answer is: Nonfluent variant primary progressive aphasia (nfvPPA)

Question 23

Not answered

Marked out of 1.00

Which of the following is NOT correct with regard to body language during a psychiatric interview?

Select one:

- Reduced eye contact may suggest passive behaviour
- Sitting on the edge of the chair may suggest anxiety
- Lack of congruent facial expression can occur in schizophrenia
- Direct eye contact suggests aggressiveness
- Lack of voice modulation can be seen in depression

Your answer is incorrect.

Reduced eye contact may suggest passive behaviour with a lack of assertiveness, but direct eye contact does not necessarily mean aggressiveness. Eye contact is also subject to the cultural background of the therapist and the patient.

The correct answer is: Direct eye contact suggests aggressiveness

Question 24

Not answered

Marked out of 1.00

A characteristic magnetic gait is seen in which of the following disorders?

Select one:

- Fronto-temporal dementia
- Parkinson disease
- Lewy body dementia
- Normal pressure hydrocephalus
- Corticobasal degeneration

Your answer is incorrect.

Explanation:

The gait dysfunction in normal pressure hydrocephalus has traditionally been described as that of a 'magnetic' or 'glue-footed' gait, gait apraxia, or a frontal ataxia similar to what has been described as Bruns ataxia. However, the gait abnormality can be highly variable and likely depends on the nature of the specific cortical-basal ganglia-thalamocortical circuitry disruption. The diagnosis should not be excluded if a magnetic gait is not seen.

Ref: Graff-Radford NR, Jones DT. Normal pressure hydrocephalus. Continuum: Lifelong Learning in Neurology. 2019 Feb 1;25(1):165-86.

The correct answer is: Normal pressure hydrocephalus

Question 25

Not answered

Marked out of 1.00

Which of the following terminology is used to refer to a psychotic patient feeling uncomfortable and perplexed?

Select one:

- Delusional mood
- Primary delusions
- Delusional perception
- Secondary delusions
- Delusional memory

Your answer is incorrect.

Explanation:

Delusional mood is recalled as arising following a period when there is an abnormal mood state characterised by anticipatory anxiety, a sense of 'something is about to happen' and an increased sense of the significance of minor events. The development of the formed delusion may come as a relief to the patient in this situation.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 107.

The correct answer is: Delusional mood

Question 26

Not answered

Marked out of 1.00

"When I saw the man holding a glass of wine, I knew my father was dead". This is an example of

Select one:

- Reflex hallucination
- Functional hallucination
- Delusional perception
- Delusional mood
- Delusional memory

Your answer is incorrect.

Explanation:

A delusional perception is a primary delusion which is recalled as having arisen as a result of a perception (e.g. a patient who, on seeing two white cars pull up in front of his house, became convinced that he was therefore about to be wrongly accused of being a paedophile). The percept is a real external object, not a hallucinatory experience. Schneider emphasised the importance of a delusional perception's 'two memberedness', as there is a link from the perceived object to the subject's perception of this object, and a second link to the new significance of the perception.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 107.

Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. pp. 48-49.

The correct answer is: Delusional perception

Question 27

Not answered

Marked out of 1.00

A patient says 'I went to the mental health centre. Centre is the gravity. Newton discovered gravity. Gravity stays with the apple. My psychiatrist took the apple away. Psychiatrists and Pediatricians are doctors'. The speech disturbance noted here is best described as

Select one:

- Loosening of association
- Flight of ideas
- Derailment
- Verbigeration
- Vorbeireden

Your answer is incorrect.

Explanation:

In flight of ideas thoughts follow each other rapidly; there is no general direction of thinking; and the connections between successive thoughts appear to be due to chance factors which, however, can usually be understood. The patient's speech is easily diverted by external stimuli and internal superficial associations. The absence of a determining tendency to thinking allows the associations of the train of thought to be determined by chance relationships, verbal associations of all kinds, clang associations, proverbs, maxims, and cliches. The chance linkage of thoughts in flight of ideas is demonstrated by the fact that one could completely reverse the sequence of the record of a flight of ideas, and the progression of thought would be understood just as well (or just as poorly).

Ref: Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. p. 44.

The correct answer is: Flight of ideas

Question 28

Not answered

Marked out of 1.00

Which of the following statements describes a characteristic feature of primary delusions?

Select one:

- Occurs secondary to auditory hallucinations
- Not preceded by delusional mood
- Primary delusions can be understood as having arisen in response to other psychiatric conditions
- Characteristically persecutory
- Frequently systematised to secondary delusions

Your answer is incorrect.

Explanation:

Primary delusions are the direct result of psychopathology, while secondary delusions can be understood as having arisen in response to other primary psychiatric conditions. Jaspers maintained that primary delusions were typical of un-understandable phenomena, being immediately experienced by the patients as absolutely self-evident like an intuition or a revelation and accompanied by abnormalities in the form such as radical transformation or emergence of thinking. He designated primary delusions as delusion proper, from which secondary delusions evolve via conscious thought processes.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 106.

Hayashi N et al. Delusion progression process from the perspective of patients with psychoses: A descriptive study based on the primary delusion concept of Karl Jaspers. Plos one. 2021 Apr 27;16(4):e0250766.

The correct answer is: Frequently systematised to secondary delusions

Question 29

Not answered

Marked out of 1.00

A psychotic patient heard hallucinatory voices only when water flowed from a tap. This phenomenon is known as a

Select one:

- Hygric hallucination
- Reflex hallucination
- Delusional misinterpretation
- Delusional perception
- Functional hallucination

Your answer is incorrect.

Explanation:

In a functional hallucination, an auditory stimulus causes a hallucination, but the stimulus as well as the hallucination is experienced. In other words, the hallucination requires the presence of another real sensation. (For example, a patient with schizophrenia first heard the voice of God as her clock ticked; later, she heard voices coming from the running tap and from the chirruping of the birds.) Both the voices and the noises are audible. Patients can distinguish both features from each other and crucially, the hallucination does not occur without the stimulus.

Ref: Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. p. 35.

The correct answer is: Functional hallucination

Question 30

Not answered

Marked out of 1.00

Which of the following is more common in mania than schizophrenia?

Select one:

- Passivity symptoms
- Persecutory delusions
- Tangentiality
- Grandiose delusions
- Circumstantiality

Your answer is incorrect.

Explanation:

Grandiose delusions are most commonly associated with manic psychosis in the context of bipolar affective disorder.

Ref: Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. pp. 51-52.

The correct answer is: Grandiose delusions

Question 31

Not answered

Marked out of 1.00

Formication is a type of

Select one:

- Deep hallucination
- Visceral hallucination
- Delusional infestation
- Haptic hallucination
- Kinaesthetic hallucination

Your answer is incorrect.

Explanation:

A haptic hallucination is a hallucination of touch. Formication is a form of tactile (or haptic) hallucination in which there is the sensation of numerous insects crawling over the surface of the body. This is not uncommon in acute organic states, including alcohol or drug withdrawal, particularly from cocaine.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1140.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 111.

Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. p. 33.

The correct answer is: Haptic hallucination

Question 32

Not answered

Marked out of 1.00

What defense mechanisms are commonly employed by patients with borderline personality disorder?

Select one:

- Repression and dissociation
- Projection and denial
- Introjection and projection
- Repression and denial
- Projective identification and splitting

Your answer is incorrect.

Explanation:

Functionally, patients with borderline personality disorder distort their relationships by considering each person to be either all good or all bad. They see people as either nurturing attachment figures or as hateful, sadistic figures who deprive them of security needs and threaten them with abandonment whenever they feel dependent. As a result of this **splitting**, the good person is idealised and the bad person devalued.

Projective identification is another defense mechanism that often occurs in patients with borderline personality disorder. In this primitive defense mechanism, the patient projects intolerable aspects of themselves onto another person, inducing them to play the projected role, and the two persons act in unison.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 564.

The correct answer is: Projective identification and splitting

Question 33

Not answered

Marked out of 1.00

A 34 year old man who has a significant conflict with his partner has the following thoughts in the unconscious and the conscious level. I hate him (unconscious); I love him (conscious). This refers to which of the following defense mechanisms?

Select one:

- Reaction formation
- Paranoia
- Sublimation
- Projection
- Denial

Your answer is incorrect.

Explanation:

Reaction formation involves externally expressing attitudes and behaviours that are the opposite of the unacceptable internal impulses (e.g. being extra polite to a person to avoid expressing anger towards them). Thus, in reaction formation, 'I hate him' becomes 'I love him.'

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 893.

The correct answer is: Reaction formation

Question 34

Not answered

Marked out of 1.00

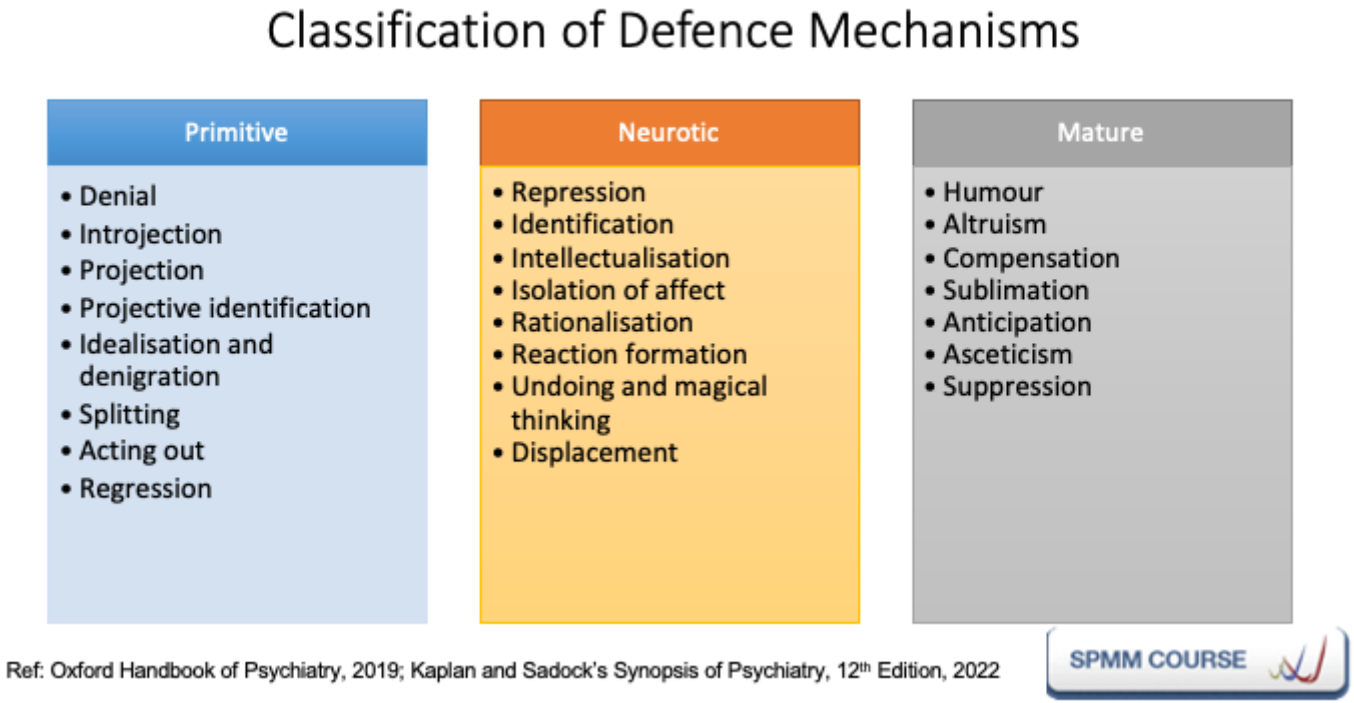
Which of the following can be considered as a neurotic defence mechanism?

Select one:

- Denial
- Altruism
- Humour
- Anticipation
- Reaction formation

Your answer is incorrect.

Explanation:



Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 892-893.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1066-1067.

The correct answer is: Reaction formation

Question 35

Not answered

Marked out of 1.00

You arrived late to a psychotherapy meeting and the therapist talks about 'Loss of ego boundaries'. Which of the following condition is she most likely referring to?

Select one:

- Depression
- Schizophrenia
- Dementia
- Borderline personality disorder
- Severe anxiety

Your answer is incorrect.

Explanation:

Patients with schizophrenia often experience their 'self' as disembodied, and they feel empty, unreal, dead, and differentiated from the world. They are unable to sustain a sense of self as persons, which is why they are equally unable to experience neither separateness from nor relatedness to others in a usual way. They may fear losing their identity in a relationship as well as feel dependent on the other for their very being. The capacity for reality testing and the relative intact ego boundaries in patients with borderline personality organisation are what delimit these patients from the psychoses.

Ref: Zandersen M, Parnas J. Identity disturbance, feelings of emptiness, and the boundaries of the schizophrenia spectrum. Schizophrenia Bulletin. 2019 Jan 1;45(1):106-13.

The correct answer is: Schizophrenia

Question 36

Not answered

Marked out of 1.00

Which of the following strategies are employed in order to reduce groupthink?

Select one:

- Not revealing the roles of individuals within the group
- Appointing a 'devil's advocate'
- Focusing on individual successes within the group
- Preventing discussion in order to save time
- Allowing decisions to be made without criticism

Your answer is incorrect.

Explanation:

Groupthink is a collective pattern of defensive avoidance, characteristic of group decision making in organisations in which group members develop rationalisations supporting shared illusions of their own infallibility and invulnerability with the organisation. Characteristic features include: (1) An illusion of invulnerability; (2) Collective efforts to rationalise in order to discount warnings; (3) An unquestioned belief in the group's inherent morality; (4) Stereotyped views of rivals and enemies; (5) Direct pressure on any member who dissents from the consensus view; (6) Self-censorship of disagreements with the apparent group consensus; (7) A shared illusion of unanimity; and (8) The emergence of self-appointed protectors of the group from adverse information that might undermine the existing complacency. Appointing a 'devil's advocate' is one strategy that can be employed to reduce groupthink.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Appointing a 'devil's advocate'

Question 37

Not answered

Marked out of 1.00

Anthony Andrews is a 7-year-old boy who has been displaying some challenging behaviours at school. His teacher produces a star chart with a star sticker being awarded to Anthony for good behaviour. Which of the following forms of conditioning would this best fit with?

Select one:

- Operant conditioning
- Classical conditioning
- Observational learning
- Imprinting
- Social conditioning

Your answer is incorrect.

Explanation:

Behaviour modification is based on operant conditioning. Forms of behavioural modification include the use of secondary reinforcement, such as 'token economy', in which socially desirable/acceptable behaviours are rewarded with tokens that can be exchanged for other material items or privileges, or 'star charts' where children's good behaviour is rewarded when a certain level is achieved.

In classical conditioning, stimuli that signal significant events produce reflexive behaviour. Observational learning is learning that occurs through observing the behaviour of others. Social conditioning is the sociological process of training individuals in a society to respond in a manner generally approved by the society in general and peer groups within society. Imprinting describes the phase-sensitive period of a young animal attaching to an external stimulus, usually its mother.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 908.

The correct answer is: Operant conditioning

Question 38

Not answered

Marked out of 1.00

You are the junior doctor at a unit for severe personality disordered patients, otherwise known as a therapeutic community. You observe group therapy and attempt to identify group dynamics based on what you have read in theory. Which of the following best describes 'group cohesion'?

Select one:

- When the group becomes self-evolving and is able to work through conflicts
- When the group works together towards a common goal
- The quality of the relationship that develops between each individual member and the therapist
- When an individual incorporates the characteristics and qualities of the group
- The process by which mere expression of ideas and conflicts is accompanied by an emotional response which produces a sense of relief

Your answer is incorrect.

Explanation:

Therapeutic factors in group psychotherapy:

Therapeutic factor	Definition
Abreaction	Repressed material, particularly a painful experience, is brought back to consciousness
Acceptance	Feeling of being accepted by other members of the group
Altruism	Act of one member helping another; putting another's needs before one's own
Catharsis	Expression of ideas/thoughts/suppressed material that is emotional and produces relief
Cohesion	Sense that the group is working together toward a common goal
Consensual validation	Confirmation of reality by comparing own conceptualisations with other group members
Contagion	Expression of emotion by a member stimulates awareness of similar emotion in another
Corrective familial experience	Group recreates the family of origin to work through original conflicts psychologically
Empathy	Capacity of group members to put themselves in psychological frame of another member
Identification	Unconscious defense mechanism; characteristics of another person are incorporated
Imitation	Conscious emulation/modeling of one's behaviour after another; 'role modeling'
Insight	Conscious awareness/understanding of own psychodynamics and maladaptive behaviour
Inspiration	Process of imparting a sense of optimism to group members; 'instillation of hope'
Interaction	Free and open exchange of ideas and feelings among group members
Interpretation	Group leader formulates the meaning of a patient's resistance, defense, and symbols
Learning	Acquiring knowledge about new areas, such as social skills; receiving advice and obtaining guidance
Reality Testing	Ability of the person to evaluate objectively the world outside the self
Transference	Projection of feelings, thoughts, and wishes onto the therapist
Universalisation	Awareness of patients that they are not alone in having problems; 'self-disclosure'
Ventilation	Expression of suppressed feelings, ideas, or events to other group members
	Adapted from Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition, 2022. p. 757.

SPMM COURSE

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 757.

The correct answer is: When the group works together towards a common goal

Question 39

Not answered

Marked out of 1.00

Cognitive Estimates Test is used to test the function of which of the following regions of the brain?

Select one:

- Dorsolateral prefrontal cortex
- Occipital Cortex
- Orbitofrontal Cortex
- Olfactory Cortex
- Anterior cingulate cortex

Your answer is incorrect.

The Cognitive Estimation Test (CET) is among a number of tests examining executive function. The dorsolateral prefrontal cortex (DLPFC), anterior cingulate cortex (ACC) and orbitofrontal cortex (OFC) all make up the prefrontal region of the frontal lobe and are all associated with executive functioning but the DLPFC is associated with, among others, problem-solving and abstract reasoning required in the CET. The ACC is associated with decision-making, motivation and response inhibition and the OFC is associated with impulse control and socially appropriate behaviours.

The correct answer is: Dorsolateral prefrontal cortex

Question 40

Not answered

Marked out of 1.00

Amy attends outpatient clinic complaining of discharge from her nipples. Which of the following medications is least associated with this side effect?

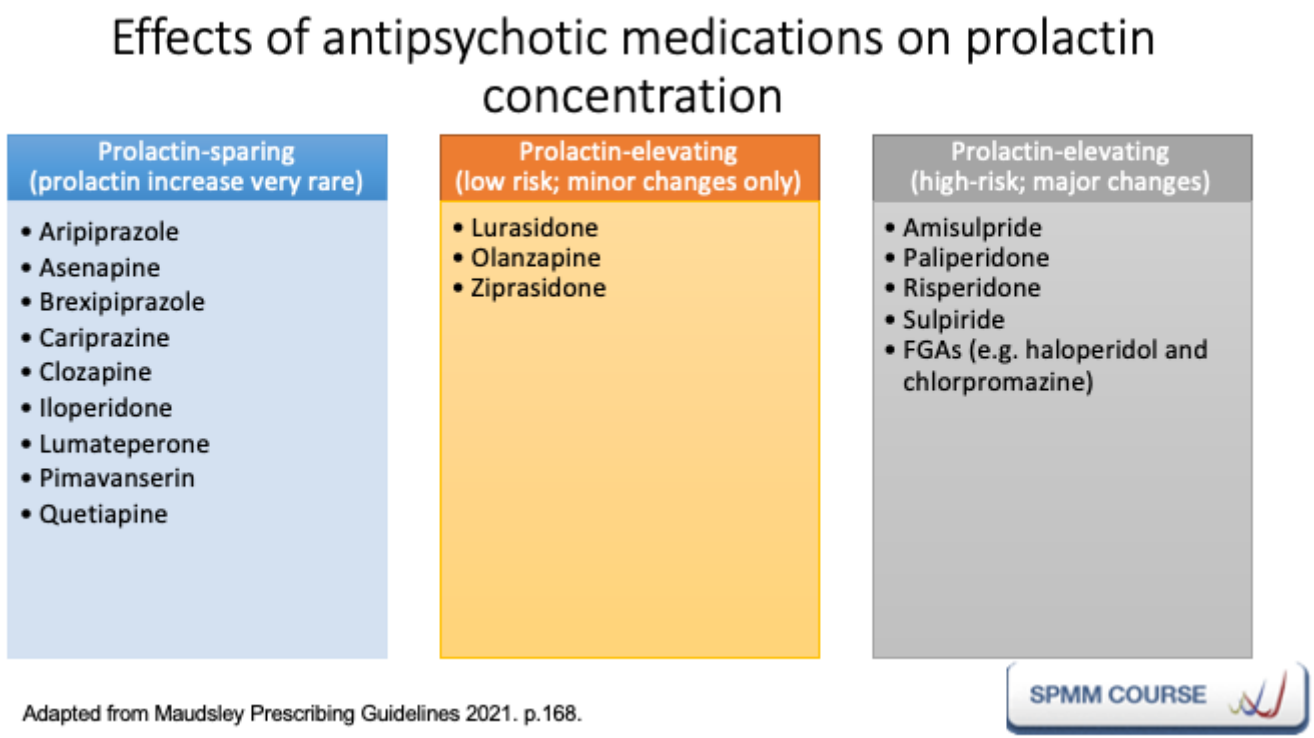
Select one:

- Sulpiride
- Haloperidol
- Amisulpride
- Risperidone
- Clozapine

Your answer is incorrect.

Explanation:

A side effect of antipsychotics can be hyperprolactinaemia. However, clozapine and aripiprazole are rarely associated with this side effect, and aripiprazole has been shown at low doses to reduce prolactin levels. This effect is attributed to the D2 receptor partial agonism, which may mitigate the effect of other antipsychotic medications on the pituitary gland.



Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 168.

The correct answer is: Clozapine

Question 41

Not answered

Marked out of 1.00

You and your colleagues are discussing various aspects of the nature of psychiatry, its history and current predicaments. One of your colleagues begins to reference a well-known book by Thomas Szasz. In his seminal book, 'The Myth of Mental Illness', Thomas Szasz makes which of the following claims?

Select one:

- The body and mind are separate
- There is no such thing as mental illness
- All problems in living can be explained by psychochemical processes
- Mental illness is a sign of brain disease
- Symptoms are explanations for mental illness

Your answer is incorrect.

Explanation:

There were three major themes in Szasz's writings: 1) His contention that there is no such thing as mental illness; 2) His contention that individual responsibility is never compromised in those suffering from what is generally considered as mental illness; 3) His perennial interest in calling attention to the political nature of psychiatric diagnosis.

Ref: Benning TB. No such thing as mental illness? Critical reflections on the major ideas and legacy of Thomas Szasz. BJPsych bulletin. 2016 Dec;40(6):292-5.

The correct answer is: There is no such thing as mental illness

Question 42

Not answered

Marked out of 1.00

Ms Winters has just joined your team as the new manager. There are some changes that need to be made to the way the team is structured. Ms Winters invited the team to contribute their ideas on how to approach the changes. She supports each team member in influencing the process. What leadership style is this describing?

Select one:

- Participative
- Laissez-faire
- Transactional
- Transformational
- Autocratic

Your answer is incorrect.

By allowing the team members to participate in the change, the challenges can be negotiated with greater ease.

The correct answer is: Participative

Question 43

Not answered

Marked out of 1.00

Dr Abdul Khan speaks English at work and enjoys going out for fish and chips with colleagues. When he goes home he speaks Punjabi and wears traditional robes. What form of acculturation does this represent?

Select one:

- Integration
- Rejection
- Separation
- Marginalisation
- Assimilation

Your answer is incorrect.

Assimilation - When individuals adopt the cultural norms of a dominant or host culture, over their original culture. Separation - When individuals reject the dominant or host culture in favor of preserving their culture of origin. Integration - When individuals are able to adopt the cultural norms of the dominant or host culture while maintaining their culture of origin. Integration leads to, and is often synonymous with biculturalism. Marginalization - When individuals reject both their culture of origin and the dominant host culture.

The correct answer is: Integration

Question 44

Not answered

Marked out of 1.00

Miss Alison Smart attends a dinner party with friends. During this she is constantly looking around to determine if she is behaving appropriately. Which phenomena does this describe?

Select one:

- Social comparison
- Social anxiety
- Halo effect
- Social perception
- Conformity

Your answer is incorrect.

Explanation:

Social comparison is the process whereby people reduce uncertainty about their beliefs, abilities, and emotions by comparing themselves with others, especially when objective criteria of evaluation are not readily available.

Social perception is the study of how people form impressions of and make inferences about other people. Halo effect is the tendency for an impression created in one area to influence opinion in another area.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Social comparison

Question 45

Not answered

Marked out of 1.00

Group therapy is often used to treat depression and anxiety; it also has a role in personality and substance misuse disorders. Which of the following is a therapeutic aspect of group therapy?

Select one:

- Intertwining
- Universality
- Stereotyping
- Sublimation
- Projective identification

Your answer is incorrect.

Irvin Yalom proposed a number of factors, 11 of which are termed curative or therapeutic factors or 'mechanisms of change that occur through an intrinsic interplay of varied guided human experiences' . These include: instillation of hope, universality, imparting information (feedback), altruism, corrective recapitulation of the primary family group, development of socialisation techniques, imitation of adaptive behaviour, interpersonal learning, group cohesiveness, catharsis, and existential factors

Therapeutic factors in group therapy:

Yalom's Therapeutic factor	Definition
Altruism	Act of one member helping another; putting another's needs before one's own
Catharsis	Expression of ideas/thoughts/suppressed material that is emotional and produces relief
Cohesion	Sense that the group is working together toward a common goal (feelings of trust/belonging emerge)
Corrective recapitulation	Group recreates the primary family group to work through original conflicts psychologically
Interpersonal Learning	Input – gaining insight on their impact from feedback; Output – providing environment for others
Imparting information	Acquiring knowledge about new areas, such as social skills; receiving advice and obtaining guidance
Imitation	Conscious emulation/modeling of one's behaviour after another; 'role modeling'
Existential factors	Members accept responsibility for decisions
Instillation of hope	Process of imparting a sense of optimism to group members / inspiration
Socialisation	Free and open exchange of ideas and feelings among group members (fosters communication)
Universality	Awareness of patients that they are not alone in having problems; 'self-disclosure'

Adapted from Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition, 2022
De Witte et al. Frontiers in Psychology. 2021:2525.

The correct answer is: Universality

Question 46

Not answered

Marked out of 1.00

Brown and Harris identified certain social factors, which increased the risk of depression. Which of the following is not one of the vulnerability factors they described?

Select one:

- Loss of mother before 11
- Lack of confiding relationship
- Physical illness
- Having 3 or more children under 15
- Lack of employment

Your answer is incorrect.

Explanation:

The Brown and Harris model of depression holds that certain 'vulnerability factors' – namely early maternal loss, lack of a confiding relationship, greater than three children under the age of 14 at home, and lack of paid employment – can interact with 'provoking agents' to increase the risk of depression for women. More recent studies support the lack of a confiding relationship as a risk factor for depression. Physical illness can be associated with depression but Brown and Harris did not identify this.

Ref: Patten SB. Are the Brown and Harris" vulnerability factors" risk factors for depression?. Journal of Psychiatry and Neuroscience. 1991 Dec;16(5):267.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 254.

The correct answer is: Physical illness

Question 47

Not answered

Marked out of 1.00

Which of the following techniques is employed in the assessment of temporal course of improvement of symptoms in patients with both PTSD and substance misuse?

Select one:

- Alcohol Use Disorders Identification Test
- Davidson Trauma scale
- Impact of Events scale
- Paddington Alcohol scale
- Marker modelling

Your answer is incorrect.

The Impact of Events scale has been used to measure temporal improvement in patients with PTSD and alcohol dependence. This was along with the Clinician Administered PTSD scale, Civilian Mississippi scale for PTSD and Time Line follow back.

The correct answer is: Impact of Events scale

Question 48

Not answered

Marked out of 1.00

Mr Ranil Premadasa is a 42-year-old Sri Lankan man who presents to his GP with concerns around persistent fatigue and various non-specific physical complaints. He attributes his symptoms to losing semen in his urine. Which of the following conditions is he most likely to be experiencing?

Select one:

- Dhat
- Susto
- Amok
- Latah
- Koro

Your answer is incorrect.

Explanation:

Dhat syndrome has been studied in South Asia and is rooted in Ayurvedic traditions about bodily production of semen as representing an end-product of energy-demanding metabolism. Dhat is recognised by a whitish discharge in the urine assumed to be semen. This 'loss of semen' is associated with somatic symptoms (weakness, exhaustion), severe anxiety, hypochondriasis, and sexual dysfunction.

Koro is a culture-bound syndrome in which an individual has an overpowering belief that one's sex organs are retracting and will disappear, despite the lack of any true longstanding changes to the genitals. Latah, from Southeast Asia, is a condition in which abnormal behaviours result from a person experiencing a sudden shock. When surprised, the affected person typically engages in such behaviours as screaming, cursing, dance movements, and uncontrollable laughter, and will typically mimic the words or actions of those around them. Susto is a culturally specific disorder primarily occurring among Latin American cultures. It is described as a condition of being frightened and "chronic

somatic suffering stemming from emotional trauma or from witnessing traumatic experiences lived by others". "Running amok" is used to refer to the behavior of someone who, in the grip of strong emotion, obtains a weapon and begins attacking people indiscriminately, often with multiple injuries. An episode of amok may be triggered by a period of depression or highly aggressive behaviour.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 990.

The correct answer is: Dhat

Question 49

Not answered

Marked out of 1.00

Children can experience emotional distress if their parents or caregivers place conflicting demands on them that allow no right or satisfactory response. Which of the following best describes this phenomenon?

Select one:

- Doublethink
- Expressed emotion
- No win situation
- Refrigerator mother
- Double bind

Your answer is incorrect.

Explanation:

Double bind is an inescapable dilemma involving conflicting demands that allow no right or satisfactory response. An influential theory of the aetiology of schizophrenia was put forward by Gregory Bateson and colleagues in 1956, according to which schizophrenia is caused by parenting styles that create double binds for children, as when a mother complains to her son for not giving her a kiss but recoils physically whenever the child does kiss her. Empirical evidence has not been forthcoming in support of this theory.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Double bind

Question 50

Not answered

Marked out of 1.00

George Wooley is an 11-year-old boy who is trying to be more productive as now he is in secondary school. Which of the following stages of Erikson's theory of psychosocial development is he most likely to be going through?

Select one:

- Industry vs inferiority
- Intimacy vs isolation
- Identity vs role confusion
- Initiative vs guilt
- Autonomy vs shame

Your answer is incorrect.

Explanation:

Erik Erikson created an original and highly influential theory of psychological development and crisis occurring in periods that extended across the entire life cycle. Stage 4 takes place from about 5 years to about 13 years. Its focus is industry versus inferiority. The child develops industry by learning new skills and takes pride in the things made. A child unprepared for this stage of psychosocial development, either through an insufficient resolution of previous stages or by current interference, may develop a sense of inferiority and inadequacy.

Erikson's Psychosocial Stage	Age	Associated Virtue	Enduring Aspects of Identity Formation
Trust vs. mistrust	Birth-18 months	Hope	Temporal perspective vs. time confusion
Autonomy vs. shame and doubt	18 months-3 years	Will	Self-certainty vs. self-consciousness
Initiative vs. guilt	3-5 years	Purpose	Role experimentation vs. role fixation
Industry vs. inferiority	5-13 years	Competence	Apprenticeship vs. work paralysis
Identity vs. role confusion	13-21 years	Fidelity	Identity vs. identity confusion
Intimacy vs. isolation	21-40 years	Love	Self-abandonment vs. self-absorption
Generativity vs. stagnation	40-60 years	Care	Leadership and followership vs. abdication of responsibility
Integrity vs. despair	60 years-death	Wisdom	Ideological commitment vs. confusion of values

Adapted from Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition, 2022. pp. 1073-1075.



Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1071-1075.

The correct answer is: Industry vs inferiority

Question 51

Not answered

Marked out of 1.00

Mr Prashant Rubasingham is a 34-year-old Sri Lankan man who moved to the UK when he was 14 years of age. He happily takes part in Sri Lankan festivities and celebrations when with his family but is also involved with his local 5-a-side football league and goes out to pubs regularly with his colleagues. Which of the following best describes this?

Select one:

- Melting pot
- Assimilation
- Separation
- Integration
- Marginalisation

Your answer is incorrect.

Integration occurs when individuals are able to adopt the cultural norms of the dominant or host culture while maintaining their culture of origin. Integration leads to, and is often synonymous with biculturalism. Assimilation occurs when individuals adopt the cultural norms of a dominant or host culture, over their original culture. Separation occurs when individuals reject the dominant or host culture in favor of preserving their culture of origin. Marginalization occurs when individuals reject both their culture of origin and the dominant host culture.

The correct answer is: Integration

Question 52

Not answered

Marked out of 1.00

Mental health-related discrimination and stigma are global, multifaceted problems. Which of the following statements regarding approaches to reduce public stigma associated with mental health disorders is correct?

Select one:

- Evidence on the effectiveness of educational interventions in changing public stigma is strong
- Lack of contact with people with stigmatising conditions reduces distrust and fear
- Protest strategies to reduce stigma improve public opinion in favour of treating mental illnesses
- People with mental illness favour approaches that address structural and institutional discrimination over public education
- 'Time to Talk' is England's largest ever program to reduce stigma and discrimination against people with mental health issues

Your answer is incorrect.

Explanation:

Evidence is mixed on the effectiveness of educational interventions in changing public stigma in a significant and lasting way. Across a wide range of stigmatising conditions, people without the stigmatised conditions have little meaningful contact with those who have these conditions. Lack of contact with people with stigmatising conditions fosters discomfort, distrust, and fear. Protest is formal objection to negative representations of people with mental illness or the nature of these illnesses. While protest may have positive outcomes in some instances, these strategies may also trigger psychological reactance or a rebound effect in which negative public opinion is strengthened as a result of the protest. People with mental illness favour approaches that address institutional and structural discrimination over those that focus on public education. 'Time to Change' is England's largest ever program to reduce stigma and discrimination against people with mental health issues, organised by charities MIND and Rethink Mental Illness. It came to a close by early 2021.

Ref: Gronholm PC et al. Interventions to reduce discrimination and stigma: the state of the art. Social psychiatry and psychiatric epidemiology. 2017 Mar 1;52(3):249-58.

Approaches to Reducing Stigma. In Ending Discrimination Against People with Mental and Substance Use Disorders: The Evidence for Stigma Change. 2016.

The correct answer is: People with mental illness favour approaches that address structural and institutional discrimination over public education

Question 53

Not answered

Marked out of 1.00

SF 36 is used to measure which of the following?

Select one:

- For assessment of physical illness alone
- Outcome measurement
- Assess psychotic symptoms
- Quality of life scale
- For assessment of mental illness alone

Your answer is incorrect.

Explanation:

The 36-Item Short Form Health Survey questionnaire (SF-36) is a very popular instrument for evaluating Health-Related Quality of Life. The SF-36 measures eight scales: physical functioning, role physical, bodily pain, general health, vitality, social functioning, role emotional, and mental health.

Ref: Lins L, Carvalho FM. SF-36 total score as a single measure of health-related quality of life: Scoping review. SAGE open medicine. 2016 Sep 30;4:2050312116671725.

The correct answer is: Quality of life scale

Question 54

Not answered

Marked out of 1.00

Negative mental health outcomes in later life relate to childhood abuse. Which of the following statements is true about childhood trauma and later outcomes?

Select one:

- Physical abuse in childhood increases risk of suicidal attempts in adolescence but not in adulthood
- Childhood trauma has no significant effect on late-life depression with onset after age 60
- Experiencing more than one type of childhood trauma escalates the risk for later negative outcomes outcomes
- Individuals with depression are less likely to remember negative events from their childhood
- Psychological, but not physical abuse, is strongly associated with depression

Your answer is incorrect.

Explanation:

All of the various types of childhood trauma (emotional, physical, and sexual abuse, physical and emotional neglect) increase the risk of depression in adulthood (odds ratio of 3). The highest odds ratios were for emotional abuse and multiple forms of maltreatment (3.82 and 4.13, respectively). Childhood emotional neglect is the most common form of maltreatment reported by patients with depression, in terms of absolute prevalence (reported by 44% of patients). Severity of depression correlates most strongly with emotional abuse. A dose-response effect where multiple types of childhood adversity increase the risk for negative health outcomes (Nelson et al., 2017)

The modes of childhood trauma that most contribute to suicidal attempts in later life (adulthood) are physical, emotional, and sexual abuse and physical neglect (Zatti et al., 2017).

The modes of childhood trauma that most contribute to late-life depression are physical and emotional abuse and physical neglect. Sexual abuse in childhood does not have a strong association with late-life depression (Wang et al., 2022)

Retrospective reports of childhood abuse are subject to mood-congruent recall bias. Depressed subjects are more likely to remember negative events from their early life; this may inflate the strength of association between abuse and depression in clinical samples. (Radell et al., 2021)

References:

Nelson J, Klumparendt A, Doebler P, Ehring T. Childhood maltreatment and characteristics of adult depression: meta-analysis. The British Journal of Psychiatry. 2017 Feb;210(2):96-104.

Zatti C, Rosa V, Barros A, Valdivia L, Calegaro VC, Freitas LH, Ceresér KM, da Rocha NS, Bastos AG, Schuch FB. Childhood trauma and suicide attempt: A meta-analysis of longitudinal studies from the last decade. Psychiatry research. 2017 Oct 1;256:353-8.

Wang Y, Chen X, Zhou K, Zhang H. A Meta-Analysis of the Effects of Childhood Maltreatment on Elderly Depression. Trauma, Violence, & Abuse. 2022 Mar 1:15248380211073838.

Radell ML, Abo Hamza EG, Daghustani WH, Perveen A, Moustafa AA. The impact of different types of abuse on depression. Depression research and treatment. 2021 Apr 13;2021.

The correct answer is: Experiencing more than one type of childhood trauma escalates the risk for later negative outcomes outcomes

Question 55

Not answered

Marked out of 1.00

The feeling of being sexually attracted and interested in a person of a particular sex denotes which of the following?

Select one:

- Sex
- Gender role
- Sexuality
- Gender identity
- Sexual attraction

Your answer is incorrect.

Explanation:

Sexuality is the range of people to whom a person is sexually attracted (including, but not limited to, heterosexuality, homosexuality, bisexuality, etc.) and unrelated to a person's gender identity.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 509.

The correct answer is: Sexuality

Question 56

Not answered

Marked out of 1.00

Sally Beech's sister is telling you about her family history of mental illness. She believes her sister has recently been diagnosed with a personality disorder. She tells you which type. Which of the following is not a type of personality disorder recognised in the DSM-5?

Select one:

- Schizoid
- Disorganised
- Paranoid
- Obsessive-compulsive
- Narcissistic

Your answer is incorrect.

Disorganised relates to a type of attachment pattern, not to a type of personality disorder.

The correct answer is: Disorganised

Question 57

Not answered

Marked out of 1.00

Jan Marchland is a 22-year-old woman with mixed anxiety and depressive disorder, who has very recently had her first baby, Sonia who was born at full term. She has been observing Sonia's eyes closely and wonders whether everything is working as it should. What visual capacities should be expected of a newborn baby?

Select one:

- Pupillary reflex
- 20/20 Acuity
- Following objects
- Accommodation
- Convergence

Your answer is incorrect.

Explanation:

The pupil light reflex has been reported to be absent in premature infants of <30 weeks gestational age but is well developed by 35 weeks. At birth, full-term newborns can see around 8-10 inches. They can track objects within a few months and focus on a nearby face by 8 weeks. Depth perception and colour vision are generally not present until 5 months of age. Accommodation develops around 3-4 months of age.

Ref: Ikeda T et al. Pupillary size and light reflex in premature infants. Neuro-Ophthalmology. 2015 Jul 4;39(4):175-8.

The correct answer is: Pupillary reflex

Question 58

Not answered

Marked out of 1.00

John Roberts is an 18 year old who has been considering attending university but has decided not to go and stay at home and help his father on the farm. According to Marcia's theory on adolescence, which of the following states is he most likely to experiencing?

Select one:

- Identity moratorium
- Identity foreclosure
- Role confusion
- Identity achievement
- Identity dissolution

Your answer is incorrect.

Explanation:

Marcia distinguished four identity statuses, based on the amount of exploration and commitment the adolescent experiences (or has experienced). In **identity diffusion**, the adolescent has not yet made commitments regarding a specific developmental task and may or may not have explored different developmental alternatives in that domain. In **foreclosure**, the adolescent has made a commitment without exploration. In **moratorium**, the adolescent is in a state of active exploration and has made no commitment, or at best an unclear one. In **identity achievement**, the adolescent has finished a period of active exploration and has made a related commitment. Role confusion is seen in Erikson's psychosocial model.

Ref: Meeus W et al. Identity statuses as developmental trajectories: A five-wave longitudinal study in early-to-middle and middle-to-late adolescents. Journal of youth and adolescence. 2012 Aug;41(8):1008-21.

The correct answer is: Identity foreclosure

Question 59

Not answered

Marked out of 1.00

In Ainsworth's Strange Situation Experiments, if a baby was found to be Type A (Anxious Avoidant), which of the following traits could be seen as an adult?

Select one:

- Greater incidence of alcohol abuse
- More likely to be victims of bullying
- Higher degree of psychopathic behaviour
- More likely to be victims of domestic violence
- More likely to be perpetrators of bullying

Your answer is incorrect.

Explanation:

Having an insecure parent attachment may be related to greater involvement in bullying for many reasons. Insecure individuals that had anxious-avoidant attachment pattern in childhood often have lower emotion regulation abilities, maladaptive social information processing, and poorer social problem-solving skills, which can all contribute to a greater propensity for bullying. Insecure individuals also tend to form negative working models of others and relationships and behave (possibly aggressively) in accordance with this perspective.

Ref: Murphy TP et al. The influences of parent and peer attachment on bullying. Journal of child and family studies. 2017 May;26(5):1388-97.

The correct answer is: More likely to be perpetrators of bullying

Question 60

Not answered

Marked out of 1.00

According to Piaget's model of cognitive development, which of the following does NOT arise during the pre-operational stage?

Select one:

- Phenomenalistic causality
- Transductive Reasoning
- Egocentrism
- Animism
- Object permanence

Your answer is incorrect.

Explanation:

In the **sensorimotor stage** (birth to 2 years), infants learn to differentiate themselves from the world. They also become able to maintain a mental image of an object, even when it is not visible (**object permanence**).

Transductive reasoning is the term used by Piaget to denote the type of thinking characteristic of children during the pre-operational stage of development. It is so called because it focuses on concrete instances and does not follow the principle of either induction or deductive reasoning. In the pre-operational stage, children also use animistic thinking, which is the tendency to endow physical events and objects with life-like psychological attributes, such as feelings and intentions. Also, children in this stage use magical thinking, called phenomenalistic causality, in which events that occur together are thought to cause one another (e.g. bad thoughts cause accidents). Egocentrism is a term given a technical meaning by Piaget to denote a cognitive state in which a child in the pre-operational stage of development comprehends the world only from its own point of view and is unaware that other people's points of view differ from their own.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1007-1008.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Object permanence

Question 61

Not answered

Marked out of 1.00

Which of the following is TRUE regarding first-born children?

Select one:

- They receive the most attention
- They are likely to have a higher IQ
- They are less achievement driven
- They are more liberal
- They are more rebellious

Your answer is incorrect.

Explanation:

As early as age one, latter-born children score lower on cognitive assessments than their siblings, and the birth order gap in cognitive assessment increases until the time of school entry and remains statistically significant thereafter.

Ref: Lehmann JY et al. The early origins of birth order differences in children's outcomes and parental behavior. Journal of Human Resources. 2018 Jan 1;53(1):123-56.

The correct answer is: They are likely to have a higher IQ

Question 62

Not answered

Marked out of 1.00

The following is true about attachment:

Select one:

- Monotropy is an uncommon phenomenon
- Attachment behaviour does not continue into adulthood
- Clear cut attachment is evident at 12 weeks
- Secure attachment occurs in about 12 to 20 months in human development
- Attachment styles are only evident from the age of 5 years

Your answer is incorrect.

Explanation:

Clear-cut attachment occurs at 6 through 24 months of age. Attachment is initially indiscriminate but becomes increasingly specific, particularly after 7 months. Monotropy, attachment to a specific individual, is seen as the rule, although children can form other bonds and attachments in time. According to Mary Ainsworth's studies, about 65% of infants are securely attached by 24 months. Attachment behaviour continues into adulthood with the attachment figures increasingly moving to peers and, in particular, romantic partners.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1011-1012.

The correct answer is: Secure attachment occurs in about 12 to 20 months in human development

Question 63

Not answered

Marked out of 1.00

In developmental psychology, the term symbiosis refers to which of the following phenomena?

Select one:

- The dual entity of father and mother
- The dual entity of mother and caregiver
- The dual entity of mother and the infant
- The dual entity of twin siblings
- The dual entity of brother and sister

Your answer is incorrect.

Explanation:

Margaret Mahler proposed a theory to describe how young children acquire a sense of identity separate from that of their mothers'. During symbiosis (age 2-5 months), mother-infant is perceived as a single fused entity.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 892-893.

The correct answer is: The dual entity of mother and the infant

Question 64

Not answered

Marked out of 1.00

The mother of an autistic child wants to know why her son lacks empathy for other children. Which of the following concepts might best explain the fundamental defect?

Select one:

- Social Exchange Theory
- Attribution Error
- Theory Of Mind
- Learning Theory
- Face Recognition Error

Your answer is incorrect.

Explanation:

'Theory of mind' refers to people's intuitive understanding of their own and other people's minds or mental states, including beliefs and thoughts. It develops by degrees from a very early age in humans: by the age of 3 years children normally have a well-developed theory of mind but cannot yet understand that people's beliefs can be false. This develops around 4 years of age. People with autistic spectrum disorder are severely impaired in their ability to understand mental states and to appreciate how mental states govern behaviour.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Theory Of Mind

Question 65

Not answered

Marked out of 1.00

When a mother leaves an infant, the infant can become upset. To compensate and comfort for this sense of loss, an object with the attributes of the mother is required. Winnicott called this object

Select one:

- Make-shift objects
- Transformable objects
- Transitional objects
- Traditional objects
- Transcultural objects

Your answer is incorrect.

Explanation:

Winnicott developed the notion of the transitional object. Ordinarily a pacifier, blanket, or teddy bear, this object serves as a substitute for the mother during infants' efforts to separate and become independent. It provides a soothing sense of security in the absence of the mother.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1089.

The correct answer is: Transitional objects

Question 66

Not answered

Marked out of 1.00

From the matching items, identify the incorrectly matched correspondence between mother's AAI (Adult attachment interview) category and infant's SSP behaviour pattern (strange situation procedure).

Select one:

- Unresolved-disorganised
- Unresolved-ambivalent
- Preoccupied-resistant
- Autonomous-secure
- Dismissing-avoidant

Your answer is incorrect.

There is an 80% correspondence between a mother's attachment category on the AAI and her infant's attachment pattern. This indicates that a mother's care-giving behaviour towards her child is significantly influenced by her own attachment experience.

Studies have shown that mothers in the dismissing adult attachment category typically have children who exhibit an avoidant attachment pattern, mothers in the pre-occupied/ entangled adult attachment category typically have children who exhibit a resistant attachment pattern, and mothers with unresolved attachment tend to have children with disorganised or disorientated attachment pattern.

Adult attachment and infant behaviour

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Main's AAI descriptors	Ainsworth's categories
Secure-autonomous: coherent description that values attachment but being objective	• Secure
Dismissive: Normalising descriptions that are unsupported or contradicted by memories	• Avoidant (insecure)
Preoccupied: with experiences (anger, fear, overwhelmed confused and passive)	• Resistant-ambivalent (insecure)
Unresolved-Disorganised: striking lapses in the monitoring of reasoning during discourse	• Disorganised (disoriented)

Adult Attachment Categories (parents) and predicted response of their infants in Strange Situation Procedure. Adapted from Main M. Introduction to the special section on attachment and psychopathology: 2. Overview of the field of attachment. Journal of consulting and clinical psychology. 1996 Apr;64(2):237.

The correct answer is: Unresolved-ambivalent

Question 67

Not answered

Marked out of 1.00

Mr. Fowler is a 30-year-old man with a diagnosis of bipolar affective disorder. His main difficulty is episodes of depression. He has attended clinic asking about the possibility of starting lamotrigine. Which of the following is a side-effect of this medication?

Select one:

- Diarrhoea
- Sialorrhoea
- Rash
- Constipation
- Nasal congestion

Your answer is incorrect.

Explanation:

10-14% of patients receiving lamotrigine will develop rash. Most are benign. A minority may be serious/life-threatening skin reactions requiring hospitalisation (e.g. Steven-Johnson syndrome, toxic epidermal necrolysis, angio-oedema, and a rash associated with a number of systemic manifestations). Rash is most likely to occur within the first 2-8 weeks of treatment and is more likely when combined with valproate, exceeding the recommended initial dose, or rapid dose escalation.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 358.

The correct answer is: Rash

Question 68

Not answered

Marked out of 1.00

Some psychiatric medications can interfere with anticoagulants with potentially severe consequences. Which of the following combinations of medication would indicate a need for an urgent INR?

Select one:

- Rivaroxaban and olanzapine
- Warfarin and St John's Wort
- Warfarin and mirtazapine
- Rivaroxaban and fluoxetine
- Unfractionated heparin and fluoxetine

Your answer is incorrect.

Explanation:

St. John’s wort (SJW) is a potent inducer of intestinal and hepatic CYP3A4, CYP2C9, CYP2C19, CYP2E1, and intestinal p-glycoprotein. Hyperforin is responsible for this effect. The hyperforin content of SJW preparations varies 50-fold, which will result in a different propensity for drug interactions between brands. Studies have shown that SJW significantly reduces plasma concentrations of warfarin, hormonal contraceptives, digoxin, and indinavir (used in the treatment of HIV).

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 418.

The correct answer is: Warfarin and St John's Wort

Question 69

Not answered

Marked out of 1.00

An ECG for your patient Will Hatton (aged 46) shows a QTc of 480m sec. Which of the following medications has the least effect on QTc interval?

Select one:

- Quetiapine
- Clomipramine
- Ziprasidone
- Aripiprazole
- Citalopram

Your answer is incorrect.

Explanation:

Aripiprazole has the lowest effect on QTc.

Effects of antipsychotics on QTc

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No effect	Low effect	Moderate effect	High effect
• Brexipiprazole • Cariprazine • Lurasidone • Lumateperone	• Aripiprazole • Asenapine • Clozapine • Flupentixol • Fluphenazine • Perphenazine • Prochlorperazine • Olanzapine • Paliperidone • Risperidone • Sulpiride	• Amisulpride • Chlorpromazine • Haloperidol • Iloperidine • Levomepromazine • Melperone • Pimavanserin • Quetiapine • Ziprasidone	• Any intravenous antipsychotic • Pimozide • Sertindole • Any drug or combination of drugs in doses exceeding recommended maximum

Adapted from Maudsley Prescribing Guidelines, 2021. p. 143.

Antidepressant	Effect on QTc (adapted from Maudsley Prescribing Guidelines, 2021. pp. 383-5.)
Agomelatine	Single case of QTc prolongation
Bupropion	QTc shortening, but prolongation has been reported in cases of overdose
Citalopram/escitalopram	Dose-related increase in QTc
Duloxetine	Isolated reports of prolongation
Fluoxetine	No effect on QTc interval
Fluvoxamine	No significant effect on QTc
Levomilnacipran	No effect on QTc interval
Lofepramine	Can possibly prolong QTc interval at higher doses
MAOIs	Unclear but may shorten QTc interval
Milnacipran	No effect on QTc
Mirtazapine	No effect on QTc
Moclobemide	No effect on QTc interval in normal doses; prolongation in overdose
Paroxetine	No effect on QTc interval
Reboxetine	No effect on QTc
Sertraline	No effect on QTc interval at standard doses; small increase at 400mg/day
Trazodone	Can prolong QTc interval
Tricyclics	Prolongation of QTc interval and QRS interval
Venlafaxine/desvenlafaxine	Possible prolongation in overdose, but very rare
Vilazodone	No effect, even in overdose
Vortioxetine	No effect

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Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 143, 383-385.

The correct answer is: Aripiprazole

Question 70

Not answered

Marked out of 1.00

Mr. Ahmed is a 58-year-old man who has been experiencing depressive symptoms following the loss of his job four months previously. Given he has a strong family history of cardiac dysfunction, the GP arranges for a routine ECG prior to prescribing an antidepressant. There is nothing of concern on the ECG, so the GP prescribes an antidepressant. After 3 months, the patient took an overdose and the ECG now shows QTc prolongation. Of the list given, which antidepressant is most associated with prolonged QTc?

Select one:

- Moclobemide
- Sertraline
- Venlafaxine
- Reboxetine
- Mirtazapine

Your answer is incorrect.

Explanation:

Of the list given, Moclobemide is possibly arrhythmogenic in overdose and QTc prolongation can occur in overdose.

Antidepressant	Effect on QTc (adapted from Maudsley Prescribing Guidelines, 2021. pp. 383-5.)
Agomelatine	Single case of QTc prolongation
Bupropion	QTc shortening, but prolongation has been reported in cases of overdose
Citalopram/escitalopram	Dose-related increase in QTc
Duloxetine	Isolated reports of prolongation
Fluoxetine	No effect on QTc interval
Fluvoxamine	No significant effect on QTc
Levomilnacipran	No effect on QTc interval
Lofepramine	Can possibly prolong QTc interval at higher doses
MAOIs	Unclear but may shorten QTc interval
Milnacipran	No effect on QTc
Mirtazapine	No effect on QTc
Moclobemide	No effect on QTc interval in normal doses; prolongation in overdose
Paroxetine	No effect on QTc interval
Reboxetine	No effect on QTc
Sertraline	No effect on QTc interval at standard doses; small increase at 400mg/day
Trazodone	Can prolong QTc interval
Tricyclics	Prolongation of QTc interval and QRS interval
Venlafaxine/desvenlafaxine	Possible prolongation in overdose, but very rare
Vilazodone	No effect, even in overdose
Vortioxetine	No effect

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Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 383-385.

The correct answer is: Moclobemide

Question 71

Not answered

Marked out of 1.00

Dawn is a 34-year-old woman with a diagnosis of epilepsy, who you have seen in clinic because of her struggles with low mood. The neurologist makes a recommendation on adding another medication to her current regime of carbamazepine. You are aware that carbamazepine can raise plasma levels of which one of the following drugs?

Select one:

- Sodium Valproate
- Phenytoin
- Lamotrigine
- Topiramate
- Lithium

Your answer is incorrect.

Explanation:

Carbamazepine can increase plasma levels of clomipramine, phenytoin, and primidone. When carbamazepine is used concurrently, phenytoin plasma levels may increase to toxic levels. Careful titration and monitoring is advised.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 134.

The correct answer is: Phenytoin

Question 72

Not answered

Marked out of 1.00

Michael has a history of psychosis. He is currently on a renal dialysis unit as his GFR is less than 10 ml/min. He has established renal failure. Which medication should be avoided in established renal impairment?

Select one:

- Olanzapine
- Aripiprazole
- Haloperidol
- Risperidone
- Amisulpride

Your answer is incorrect.

Explanation:

Amisulpride and sulpiride are almost totally renally excreted and best avoided in renal impairment and cases with established renal failure.

Drug group	Recommended psychotropics in renal impairment
Antipsychotics	No agent clearly preferred to another, however: **Avoid** sulpiride and amisulpride **Avoid** highly anticholinergic drugs because they can contribute to urinary retention First-generation antipsychotic: suggest **haloperidol** 2-6mg/day Second-generation antipsychotic: suggest **olanzapine** 5mg/day
Antidepressants	No agent clearly preferred to another, however: **Citalopram** (care re: QTc-prolonging effects) and **sertraline** are reasonable choices **CBT** where available
Mood stabilisers	No agent clearly preferred to another, however: **Avoid** lithium if possible Suggest starting **valproate or lamotrigine** at low dose, increase slowly, and monitor for adverse effects
Anxiolytics and hypnotics	No agent clearly preferred to another, however: Excessive sedation more likely to occur, so monitor all patients carefully **Lorazepam** and **zopiclone** are reasonable choices
Anti-dementia drugs	No agent clearly preferred to another, however: Rivastigmine is a reasonable choice

Adapted from Maudsley Prescribing Guidelines 2021. p. 749.

SPMM COURSE

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 749.

The correct answer is: Amisulpride

Question 73

Not answered

Marked out of 1.00

Nadine is a 46-year-old French teacher with a diagnosis of bipolar affective disorder. You wish to commence her on lithium and enquire about her medication history to exclude possible interactions. Which of the following medications reduces lithium excretion?

Select one:

- Lamotrigine
- Sertraline
- Ramipril
- Propanolol
- Amiloride

Your answer is incorrect.

Explanation:

ACE inhibitors (such as ramipril) can reduce thirst (which can lead to mild dehydration) and increase renal sodium loss (leading to increased sodium re-absorption by the kidney), resulting in an increase in lithium plasma levels. The magnitude of this effect is variable, from no increase to a four-fold increase. The risk seems to be increased in patients with heart failure, dehydration, and renal impairment. In the elderly, ACE inhibitors increase seven-fold the risk of hospitalisation due to lithium toxicity. ACE inhibitors can also precipitate renal failure so, if co-prescribed with lithium, more frequent monitoring of eGFR and plasma lithium is required.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 252-253.

The correct answer is: Ramipril

Question 74

Not answered

Marked out of 1.00

Mark attends clinic complaining of a side effect of his medication. Which of the following describes a slow rhythmic movement disorder affecting the extremities most markedly?

Select one:

- Athetosis
- Dystonia
- Tic
- Akathisia
- Sydenham chorea

Your answer is incorrect.

Explanation:

Athetosis is characterised by involuntary, slow, writhing movements, especially of the fingers, hands, head, and tongue. Akathisia is a feeling of inner restlessness and desire to move. A tic is a habitual spasmodic contraction of muscles. Sydenham chorea is rapid uncoordinated jerky movements.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Athetosis

Question 75

Not answered

Marked out of 1.00

Miss Clarke is a 32-year-old woman with a diagnosis of opiate dependence syndrome who is under a substance misuse service. She attends clinic and sees you to request that you prescribe her an alpha-2-agonist medication to prevent her experiencing withdrawals whilst she stops using opiates. She has a history of experiencing marked hypotension in the past. Which of the following medications would be most appropriate for Miss Clarke?

Select one:

- Clonidine
- Tizanidine
- Romifidine
- Lofexidine
- Guanfacine

Your answer is incorrect.

Both Clonidine and Lofexidine are used to reduce the withdrawal symptoms associated with stopping opiates. Clonidine is commonly associated with hypotension, whilst this is not seen with lofexidine with the same frequency. Interesting, clonidine can be used in some patients with profound and persistent hypotension due to it having a limited central hypotensive effect in these patients.

The correct answer is: Lofexidine

Question 76

Not answered

Marked out of 1.00

Which of the following best describes how methylphenidate works?

Select one:

- Inhibition of the hypothalamus
- Serotonin reuptake inhibitor
- Blocking dopamine transporter
- Histamine antagonist
- Blocking orexin

Your answer is incorrect.

Explanation:

Stimulants (perhaps better designated as norepinephrine (NE) and dopamine (DA) reuptake blockers) such as methylphenidate block the transporters both for NE and DA (NETs and DATs, respectively). Methylphenidate blocks DATs and NATs allosterically, stopping the reuptake of DA via DATs and NE via NETs.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 469.

The correct answer is: Blocking dopamine transporter

Question 77

Not answered

Marked out of 1.00

When she was a teenager, Gemma struggled with her self-esteem and self-image. She exercised several times a day and severely restricted her dietary intake. Despite this, she wanted to lose further weight and bought some pills online. Her GP had also recently increased Gemma's dose of venlafaxine the previous week with the hope of helping Gemma with her feelings of low mood. Three days after taking the pills bought online, Gemma's mother found her collapsed on her bed, sweating profusely, in an altered conscious state. What medication could Gemma have ordered on the internet that may have caused this?

Select one:

- Guarana
- Saffron extract
- Orlistat
- Magnesium
- Sibutramine

Your answer is incorrect.

Sibutramine (formerly sold as Meridia) is a monoamine reuptake inhibitor, and acts as an appetite suppressant. Risks of concomitant use with SSRIs and SNRIs include serotonin syndrome, which can manifest with confusion, altered consciousness, sweating, muscle rigidity, tachycardia, dilated pupils, and required urgent medical intervention. Sibutramine has been withdrawn from the UK and many other countries.

The correct answer is: Sibutramine

Question 78

Not answered

Marked out of 1.00

Charles is a 53-year-old male with a diagnosis of moderate depressive disorder. His psychiatrist prescribed him amitriptyline two months ago. At his follow up appointment, Charles reports an improvement in his mood and sleep but complains of lightheadedness. His psychiatrist measures his sitting-standing blood pressure and notices a considerable drop in his systolic blood pressure, which satisfies the criteria for orthostatic hypotension. Which receptor blockade leads to this kind of effect?

Select one:

- Beta receptors
- D2 receptors
- 5-HT2A receptors
- Histamine
- Alpha-1 receptors

Your answer is incorrect.

Explanation:

Amitriptyline is a tricyclic antidepressant that acts as a serotonin and norepinephrine reuptake inhibitor. Its blockade of alpha adrenergic 1 receptor can cause dizziness, sedation, and hypotension. Sedative effects and weight gain can occur due to antihistaminic properties. Anticholinergic activity can cause sedative effects, dry mouth, constipation, and blurred vision.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 25-26.

The correct answer is: Alpha-1 receptors

Question 79

Not answered

Marked out of 1.00

Which of the following describes the mechanism of action of Pregabalin?

Select one:

- Potentiates effects of GABA
- Blocks glutamate
- Has a mechanism of action identical to GABA
- Binds to voltage-gated calcium channels
- Binds GABA-A sites

Your answer is incorrect.

Explanation:

Pregabalin is a leucine analog and is transported both into the blood from the gut and also across the blood-brain barrier into the brain by the system L transport system (a sodium-dependent transporter) as well as by additional sodium-dependent amino acid transporter systems. Pregabalin binds to the alpha 2 delta subunit of voltage-sensitive calcium channels. This closes N and P/Q presynaptic calcium channels, diminishing excessive neuronal activity and neurotransmitter release. Although structurally related to gamma-aminobutyric acid (GABA), it has no known direct actions on GABA or its receptors.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 647.

The correct answer is: Binds to voltage-gated calcium channels

Question 80

Not answered

Marked out of 1.00

Blocking dopamine in which of the following pathways would result in hyperprolactinaemia?

Select one:

- Cerebellar
- Mesolimbic
- Nigrostriatal
- Tuberoinfundibular
- Mesocortical

Your answer is incorrect.

Explanation:

The dopamine neurons that project from hypothalamus to anterior pituitary gland are known as the **tuberoinfundibular** dopamine pathway. Normally, these neurons are tonically active and inhibit prolactin release. In the postpartum state, however, the activity of these dopamine neurons is decreased. Prolactin levels can therefore rise during breast feeding so that lactation will occur. If the functioning of the tuberoinfundibular dopamine neurons is disrupted by lesions or drugs, prolactin levels can also rise.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 85.

The correct answer is: Tuberoinfundibular

Question 81

Not answered

Marked out of 1.00

Zopiclone acts on which receptor site?

Select one:

- Omega-1 subunit of Alpha 1 receptor
- Alpha-1 subunit of Omega 1 receptor
- Alpha-2 subunit of Omega 2 receptor
- Delta-1 subunit of Omega 1 receptor
- Omega-1 subunit of Sigma 1 receptor

Your answer is incorrect.

Explanation:

Z drugs bind to benzodiazepine-sensitive GABA-A receptors. A different nomenclature (omega system) denotes the specific sites of GABA-A receptors where these 'Z' drugs act as omega-1 receptors. Conventional benzodiazepines interact at both ω1 and ω2 sites, the latter likely contributing to side effects.

Zolpidem and zaleplon bind selectively to alpha-1 subunits of these sites. By contrast, benzodiazepines and zopiclone/eszopiclone bind to four alpha subunits (alpha-1, alpha-2, alpha-3, and alpha-5). The functional significance of alpha-1 selectivity is not yet proven, but may contribute to lower risk of tolerance and dependence. The alpha-1 subtype is known to be critical for producing sedation and thus is targeted by every effective GABA-A PAM (positive allosteric modulator) hypnotic, both benzodiazepines and Z-drugs.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 422-423.

Madari S, Golebiowski R, Mansukhani MP, Kolla BP. Pharmacological management of insomnia. Neurotherapeutics. 2021 Jan;18(1):44-52.

The correct answer is: Alpha-1 subunit of Omega 1 receptor

Question 82

Not answered

Marked out of 1.00

Mr. Zane is a 60-year-old man who is on warfarin following a DVT. Which of the following medications would be the most important for him to avoid?

Select one:

- St John's Wort
- Risperidone
- Fluoxetine
- Lamotrigine
- Lithium

Your answer is incorrect.

Explanation:

St. John's wort (SJW) is a potent inducer of intestinal and hepatic CYP3A4, CYP2C9, CYP2C19, CYP2E1, and intestinal p-glycoprotein. Hyperforin is responsible for this effect. The hyperforin content of SJW preparations varies 50-fold, which will result in a different propensity for drug interactions between brands. Studies have shown that SJW significantly reduces plasma concentrations of warfarin, hormonal contraceptives, digoxin, and indinavir (used in the treatment of HIV).

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 418.

The correct answer is: St John's Wort

Question 83

Not answered

Marked out of 1.00

Women with mental health conditions often need to continue medication during pregnancy and breastfeeding. The benefit from medicines during this time must be balanced against possible risks to the newborn and neonate. Which of the following drugs is considered to be the most teratogenic?

Select one:

- Lithium
- Dosulepin
- Carbamazepine
- Valproate
- Lamotrigine

Your answer is incorrect.

Explanation:

Valproate is associated with strong evidence for increased risk of congenital and major congenital malformations (as many as 11% of exposed children); cleft lip and/or palate, neural tube defects (risk of spina bifida 1-2%), congenital heart anomalies, radial ray defects, ophthalmological and genitourinary anomalies; developmental delay (delayed walking and talking, memory problems, difficulty with speech and language), low IQ, and autism. This risk increases with increasing doses of valproic acid, especially when the dose is more than 1gm/day.

Ref: McAllister-Williams RH et al. British Association for Psychopharmacology consensus guidance on the use of psychotropic medication preconception, in pregnancy and postpartum 2017. Journal of Psychopharmacology. 2017 May;31(5):519-52.

National Institute for Health and Clinical Excellence. Antenatal and postnatal mental health: clinical management and service guide. Updated edition 2020.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 690-691.

The correct answer is: Valproate

Question 84

Not answered

Marked out of 1.00

Miss Davies is a 28-year-old woman with a diagnosis of opiate dependence. She is keen to become abstinent and asks if there is any medication she could take that would counteract the effect of opioids and help her stop their use. Which of the following is a medication that would best fit this description?

Select one:

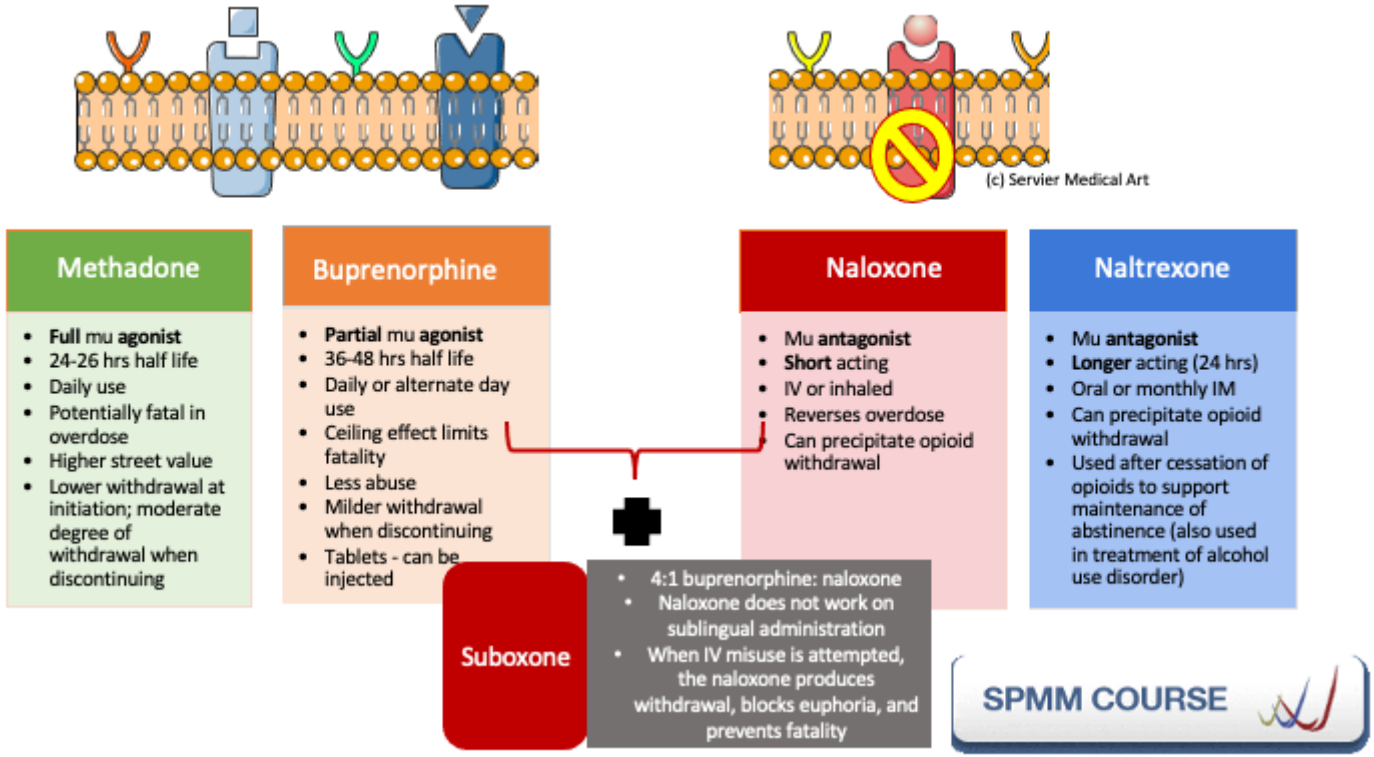
- Clonidine
- Naloxone
- Methadone
- Baclofen
- Naltrexone

Your answer is incorrect.

Explanation:

Naltrexone has been found by NICE to be a cost-effective treatment strategy in aiding abstinence from opioid misuse for those who prefer an **abstinence programme** (rather than substitute prescribing of methadone or buprenorphine), are **fully informed** of potential risks and benefits, are **highly motivated** to remain on treatment, and have a **partner supporting concordance.** It hasthe propensity to cause a **severe withdrawal reaction** from opioids. A **test dose of naloxone**, which has a much **shorter half-life** than naltrexone, may be given prior to starting naltrexone treatment; any withdrawal symptoms precipitated will be of shorter duration than if precipitated by naltrexone. **Discontinuation** may be associated with an increase in **inadvertent overdose** from illicit opioids, making **supervision** of administration and **careful choice** of who is prescribed it (**abstinence-focused and motivated**) very important. People taking naltrexone often experience **adverse effects** of unease (dysphoria), depression and insomnia, which can lead to relapse to **illicit opioid use** or **failure to continue treatment.**

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 496.



The correct answer is: Naltrexone

Question 85

Not answered

Marked out of 1.00

Which of the following neurotransmitters is primarily released from the medial and dorsal raphe nucleus?

Select one:

- Noradrenaline
- Serotonin
- Dopamine
- Acetylcholine
- GABA

Your answer is incorrect.

Explanation:

Serotonin has both ascending and descending projections. Ascending serotonergic projections originate mainly in the raphe nucleus in the brainstem. These ascending projections may regulate mood, anxiety, sleep, and other functions. Descending serotonergic projections extend down the brainstem and through the spinal cord. They may regulate pain.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 279.

The correct answer is: Serotonin

Question 86

Not answered

Marked out of 1.00

Rosalind is a 75-year-old female who presents to your clinic with a five-week history of low mood, anhedonia, poor appetite and insomnia. You diagnose her with moderate depressive disorder and wish to start her on antidepressants. You ask her about other medications she is taking and she tells you that she is taking amlodipine and warfarin. Which of the following antidepressants will you avoid prescribing?

Select one:

- Trazadone
- Mirtazapine
- Venlafaxine
- Fluoxetine
- Duloxetine

Your answer is incorrect.

Explanation:

Fluoxetine may displace highly protein-bound drugs like warfarin. Fluoxetine carries a possible increased risk of bleeding, especially when combined with anticoagulants (e.g. warfarin, NSAIDs).

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 304.

The correct answer is: Fluoxetine

Question 87

Not answered

Marked out of 1.00

Which of the following psychotropic drugs is affected by acetylator (slow/fast) status?

Select one:

- Oxazepam
- Galantamine
- Chlorpromazine
- Lithium
- Nitrazepam

Your answer is incorrect.

- Acetylation is a phase 2 nonmicrosomal conjugation reaction.
- Drugs such as isoniazid, hydralazine, procainamide, some sulfonamides, sulfasalazine, nitrazepam, and caffeine undergo acetylation.
- The enzyme group responsible are the N-acetyltransferases.
- There are two different isozymes in humans, encoded by the genes NAT1 and NAT2.
- Based on the polymorphisms of NAT1 and NAT2, humans can be fast, intermediate, or slow acetylators.
- 60% of Caucasians and 10% to 20% of Asians are slow acetylators.
- Slow acetylation contributes to prolonged half-life of the drug. This increases the risk of side effects such as peripheral neuropathy (isoniazid), lupus syndrome (hydralazine and procainamaide), allergic reactions and hemolysis (sulfonamides), drowsiness, hangover (nitrazepam), and gastrointestinal effects (sulfasalazine).

- Ref: Kant A, Bilmen J, Hopkins PM. Adverse drug reactions. In Pharmacology and Physiology for Anesthesia 2019 Jan 1 (pp. 130-143). Elsevier.

The correct answer is: Nitrazepam

Question 88

Not answered

Marked out of 1.00

Mrs. X suffers from depressive disorder and has been on paroxetine for the last 8 weeks. She has not made any improvement and stopped taking her medication. She is now complaining of dizziness, insomnia and shock like sensations. The most likely diagnosis is

Select one:

- SSRI related toxicity
- Serotonin syndrome
- Neuroleptic malignant syndrome
- Teratogenicity
- SSRI discontinuation syndrome

Your answer is incorrect.

Explanation:

SSRI discontinuation symptoms may include flu-like symptoms, 'shock-like' sensations, dizziness (most common), insomnia, excessive dreaming, irritability, weakness, lethargy/fatigue, GI symptoms, and tearfulness. The onset of symptoms is usually within days (or 3-5 drug half-lives) of stopping treatment.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 143.

The correct answer is: SSRI discontinuation syndrome

Question 89

Not answered

Marked out of 1.00

Julie is a 29-year-old woman who has recently started taking combined oral contraceptive pills. Which of the following medication could decrease the effectiveness of the contraception?

Select one:

- St. John's wort
- Fluoxetine
- Lithium
- Clozapine
- Amitriptyline

Your answer is incorrect.

Explanation:

Psychiatric drugs that are CYP3A4 inducers (e.g. carbamazepine, St John's Wort, phenytoin, topiramate) reduce the bioavailability and hence contraceptive efficacy of combined hormonal contraceptives, the progesterone-only pill, and the subdermal implant both during use and for up to 28 days after stopping enzyme-inducer treatment. Thus, patients on enzyme inducers should be encouraged to use intrauterine contraception or injectables.

Ref: Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. p. 314.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 855.

The correct answer is: St. John's wort

Question 90

Not answered

Marked out of 1.00

A 69-year-old man on warfarin has come back from a holiday in France when he developed some psychiatric symptoms. He was also taking one of the following drugs. Which one of the above drugs would lead you to send him for an urgent INR?

Select one:

- Clozapine
- St. John's wort
- Olanzapine
- Lithium
- Venlafaxine

Your answer is incorrect.

Explanation:

St. John's wort (SJW) is a potent inducer of intestinal and hepatic CYP3A4, CYP2C9, CYP2C19, CYP2E1, and intestinal p-glycoprotein. Hyperforin is responsible for this effect. The hyperforin content of SJW preparations varies 50-fold, which will result in a different propensity for drug interactions between brands. Studies have shown that SJW significantly reduces plasma concentrations of warfarin, hormonal contraceptives, digoxin, and indinavir (used in the treatment of HIV).

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 418.

The correct answer is: St. John's wort

Question 91

Not answered

Marked out of 1.00

Which of the following is correct with respect to first-order elimination kinetics?

Select one:

- Constant amount of drug is eliminated per unit of time
- Very few drugs undergo first order metabolism
- Steady state concentration of the drug is doubled by doubling the administration rate
- Clearance depends on concentration of the drug in plasma
- Most commonly seen when metabolic enzymes are saturated

Your answer is incorrect.

Explanation:

Constant fraction, not amount is eliminated per unit time if a drug undergoes first order elimination. Hence, clearance is independent of drug concentration in plasma. Doubling the administration rate will double the steady-state plasma concentration.

Ref: Laracuente ML, Marina HY, McHugh KJ. Zero-order drug delivery: State of the art and future prospects. Journal of Controlled Release. 2020 Nov 10;327:834-56.

The correct answer is: Steady state concentration of the drug is doubled by doubling the administration rate

Question 92

Not answered

Marked out of 1.00

Which of the following statements about protein binding is incorrect?

Select one:

- Tissue binding of a drug cannot be directly measured
- The distribution of a drug is not affected by the degree of protein binding
- Plasma protein binding is usually reversible
- Pregnancy can affect protein binding of drugs
- Displacement bound fraction of a drug from plasma proteins may be a basis of drug interactions.

Your answer is incorrect.

The distribution of a drug depends on how protein bound it is. When in the blood, many drugs are bound to circulating plasma proteins, predominantly albumin but also globulins and glycoproteins. It is the unbound fraction of the drug (free fraction) that can be active i.e., bind to receptors, pass across blood brain barrier, etc. Therefore, changes in protein binding can have profound effects on the availability of the drugs. Most psychotropic drugs are basic, and they may bind to, for example, alpha-1 acid glycoprotein and lipoproteins. Plasma protein binding is usually reversible (not covalent).

The correct answer is: The distribution of a drug is not affected by the degree of protein binding

Question 93

Not answered

Marked out of 1.00

Mr. Brown is a retired biologist and has recently been diagnosed with a neurodegenerative condition. He is interested to know what particular proteins have been damaged in his brain. He looks up some information on hyperphosphorylation of tau protein. Which one of the following conditions is not considered to be a tauopathy?

Select one:

- Corticobasal degeneration
- Pick disease
- Progressive supranuclear palsy
- Korsakoff syndrome
- Alzheimer dementia

Your answer is incorrect.

Explanation:

Tauopathies including but not limited to Alzheimer disease, some frontotemporal dementias (e.g. Pick disease), corticobasal degeneration, progressive supranuclear palsy, and chronic traumatic encephalopathy, are progressive neurodegenerative disorders that are pathologically defined by tau-positive deposits in the brain.

Ref: Orr ME et al. A brief overview of tauopathy: causes, consequences, and therapeutic strategies. Trends in pharmacological sciences. 2017 Jul 1;38(7):637-48.

The correct answer is: Korsakoff syndrome

Question 94

Not answered

Marked out of 1.00

Mr Harry Lime is a 69-year-old man who presents to A&E with a left-sided facial palsy and right upper limb and lower limb weakness. In which of the following areas is the causative lesion most likely to be present?

Select one:

- Right pons
- Cortex
- Left mid brain
- Right mid brain
- Left pons

Your answer is incorrect.

In pontine stroke, patients often experience ipsilateral facial palsy and contralateral limb paresis. In midbrain strokes, patients often experience contralateral facial and limb paresis but ipsilateral cranial nerve III palsy.

The correct answer is: Left pons

Question 95

Not answered

Marked out of 1.00

Mr Jones is a 65-year-old man who has been seen by his GP for disturbed sleep. He has recently begun to thrash around at night, punch and kick and has occasionally injured his wife. He has read about such sleep problems and is worried he may be at higher risk of developing a neurological condition in the future. Which one of the following neurological conditions has not been associated with REM sleep behaviour disorder?

Select one:

- Multiple system atrophy
- Frontotemporal dementia
- Lewy body dementia
- Parkinson's disease
- Mononeuritis complex

Your answer is incorrect.

Alpha-synuclein pathology, as seen in Parkinson disease (PD), multiple system atrophy (MSA), and dementia with Lewy bodies (DLB), is the most common underlying cause of RBD (REM Sleep Behaviour Disorder). Less commonly, RBD occurs in non-synuclein neurodegenerative disorders such as progressive supranuclear palsy, frontotemporal dementia, amyotrophic lateral sclerosis, Alzheimer disease, spinal cerebellar ataxia type 3, Huntington disease, and myotonic dystrophy type. Mononeuritis multiplex has not been associated with this illness. (Ref: Chan PC, Lee HH, Hong CT, Hu CJ, Wu D. REM Sleep Behavior Disorder (RBD) in Dementia with Lewy Bodies (DLB). Behavioural Neurology. 2018)

The correct answer is: Mononeuritis complex

Question 96

Not answered

Marked out of 1.00

Mr. Ronald White is a 50-year-old man who has experienced a traumatic brain injury. He was discharged from hospital following this but has now been readmitted due to repeated episodes of anger outbursts. His wife reports that his memory appears to be gradually worsening. Which of the following tests would be most useful to clarify the diagnosis?

Select one:

- IPDE
- MRI head
- Wisconsin card sorting test
- Myers-Briggs test
- MMPI

Your answer is incorrect.

An MRI head to check for intracranial complications of the initial injury would be the most sensible approach. The Wisconsin card test tests an individual's ability to “set shift" between one task and another. The International personality disorder examination, Minnesota Multiphasic Personality Inventory and Myers-Briggs Personality Inventory are unlikely to be appropriate for an individual with a brain injury and declining cognition.

The correct answer is: MRI head

Question 97

Not answered

Marked out of 1.00

Mr. Howard James is a 73-year-old man who has recently experienced a stroke. His brother who is nursing him at the hospital has observed that Howard fails notice people on one side of his bed. A lesion in which of the following areas is most likely to lead to this presentation?

Select one:

- Occipital lobe
- Dominant temporal lobe
- Non-dominant temporal lobe
- Dominant parietal lobe
- Non-dominant parietal lobe

Your answer is incorrect.

Explanation:

Neglect syndromes are characterised by failure to detect visual or tactile stimuli or to move the limb in the contralateral hemispace. They are most associated with right hemisphere damage in the parietal region. The right parietal lobe is non-dominant in most individuals.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 49.

The correct answer is: Non-dominant parietal lobe

Question 98

Not answered

Marked out of 1.00

Sleep terrors most commonly occur during which of the following stages of sleep?

Select one:

- Fast wave sleep
- REM
- Non-REM stage 2
- Non-REM stage 1
- Non-REM stage 3

Your answer is incorrect.

Explanation:

Sleep terrors are recurrent episodes of **abrupt awakening** from sleep characterised by a panicky scream, with intense fear and autonomic arousal. The individual usually has **no recall** of the details of the event and is **unresponsive to stimuli** (e.g conversation) during the episode. **Autonomic hyperactivity** is manifested by tachycardia, tachypnea, diaphoresis, flushed face, dilated pupils, agitation, tremulousness, and increased muscle tone. A typical episode **lasts no more than a few minutes**, but may be protracted, lasting up to an hour.

Sleep terrors occur during the **first third of the night**, during **slow-wave stage 3 of NREM sleep.** NREM is classified as stages 1 to 3 with increasing amplitude and decreasing frequency of EEG activity. Stage 3 (previously split as 3 & 4) constitutes slow wave sleep (SWS). SWS dominates initial part of the sleep.

Ref: Leung AK et al. Sleep terrors: an updated review. Current pediatric reviews. 2020 Aug 1;16(3):176-82.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 462.

The correct answer is: Non-REM stage 3

Question 99

Not answered

Marked out of 1.00

Neurofibrillary tangles (NFT) can be detected via the use of labelled antibodies, immunostaining, which attach to specific molecules within their structure. Which of the following immunostaining findings are associated with classical neurofibrillary tangles?

Select one:

- Tau negative
- Alpha-synuclein positive
- Amyloid positive
- Cystatin C positive
- Tau positive

Your answer is incorrect.

Explanation:

Neurofibrillary tangles are intracellular inclusion bodies, primarily containing tau as well as other proteins. Tau is a microtubule-binding protein, but in tangles, it has separated from the microtubules and is hyperphosphorylated at a variety of serine and threonine residues. Tangles may be detected by immunostaining for tau protein, as well as silver impregnation techniques such as Bielschowsky or Gallyas stains. Tangles are particularly well demonstrated by tau immunocytochemistry, which is now used routinely in diagnostic work. Neurofibrillary tangles may on occasion also be immunoreactive for ubiquitin and p62.

Ref: Gray F, Duyckaerts C, de Girolami U. Escourolle and Poirier's Manual of Basic Neuropathology, 6th Edition. 2019. pp. 5, 189.

The correct answer is: Tau positive

Question 100

Not answered

Marked out of 1.00

The brains of patients with schizophrenia can show some structural differences compared to those of controls. Which of the following brain areas tend to be larger in patients with schizophrenia?

Select one:

- Ventricles
- Medial temporal lobes
- Superior temporal lobes
- Nucleus accumbens
- Prefrontal cortex

Your answer is incorrect.

Explanation:

Neuropathology in schizophrenia includes larger ventricles, decreases in gray matter and whole-brain volume, white matter anisotropy, reduced prefrontal lobe volume, and reduced volume of the medial temporal lobes.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 358.

Hollis C. Schizophrenia in children and adolescents. BJPsych Advances. 2015 Sep;21(5):333-41.

Bakhshi K, Chance SA. The neuropathology of schizophrenia: a selective review of past studies and emerging themes in brain structure and cytoarchitecture. Neuroscience. 2015 Sep 10;303:82-102.

The correct answer is: Ventricles

Question 101

Not answered

Marked out of 1.00

Impaired odour identification is most likely to develop in which of the following conditions?

Select one:

- CJD
- Multiple sclerosis
- Vascular dementia
- Depression
- Alzheimer disease

Your answer is incorrect.

Explanation:

Odour identification ability is impaired in patients with Alzheimer disease (AD). It remains unclear whether AD pathology directly affects odour identification ability, or if it is mediated by cognitive dysfunction.

Ref: Baek MS et al. Effect of A/T/N imaging biomarkers on impaired odor identification in Alzheimer's disease. Scientific reports. 2020 Jul 14;10(1):1-9.

The correct answer is: Alzheimer disease

Question 102

Not answered

Marked out of 1.00

Which of the following brain regions plays an important role in the memory of emotionally salient events?

Select one:

- Anterior cingulate cortex
- Amygdala
- Nucleus basalis
- Insula
- Locus ceruleus

Your answer is incorrect.

Explanation:

The amygdala is an almond-shaped brain structure in the limbic system at the base of the inside of each temporal lobe, contiguous with the olfactory cortex, controlling the experience and expression of emotion and involved in motivation, aggression, feeding, and (through its links with the hippocampus) long-term memory. Electrical stimulation of this area usually produces an intense emotion of fear.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Amygdala

Question 103

Not answered

Marked out of 1.00

Damage to which part of the brain can result in Anton's syndrome?

Select one:

- Cingulate gyrus
- Fusiform gyrus
- Superior temporal gyrus
- Non-dominant parietal lobe
- Bilateral Occipital lobes

Your answer is incorrect.

Anton–Babinski syndrome, also known as visual anosognosia, is a rare symptom of brain damage occurring in the occipital lobe. Those who suffer from it are "cortically blind", but insist that they are capable of seeing despite evidence to the contrary. Sufferers dismiss evidence of their condition and employ confabulation to fill in the missing sensory input.

The correct answer is: Bilateral Occipital lobes

Question 104

Not answered

Marked out of 1.00

The form of Down syndrome which is as likely to have come from the father's as that of mother's chromosome is:

Select one:

- Mosaicism
- Duplication
- Robertsonian translocation
- Deletion
- Full trisomy 21

Your answer is incorrect.

Explanation:

Partial or complete trisomy 21 (that is, the presence of part of or a complete supernumerary chromosome 21) is the genomic cause of Down syndrome (DS). Most cases of DS are not inherited. However, people with translocation DS can inherit the condition from an unaffected parent (either paternal or maternal). Translocation accounts for trisomy 21 in ~5% of affected individuals, usually t(14;21) or t(21;21). In translocation, a fusion occurs, of two chromosomes, resulting in a total of 46 chromosomes, despite the presence of an extra chromosome 21. This is usually inherited, and the translocated chromosome may be found in unaffected parents and siblings. Asymptomatic carriers have only 45 chromosomes.

A free trisomy 21 is present in 95% of individuals with DS and results from an error in maternal meiosis I (~66%) or meiosis II (~21%); paternal meiosis I (~3%) or meiosis II (~5%); or mitosis, after the formation of the zygote (5%). Advanced maternal age has been associated with chromosome 21 segregation errors in both maternal meiosis I and meiosis II. Mosaicism for trisomy 21 is not inherited and occurs in ~2% of individuals with DS.

Ref: Antonarakis SE et al. Down syndrome. Nature Reviews Disease Primers. 2020 Feb 6;6(1):1-20.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 101.

The correct answer is: Robertsonian translocation

Question 105

Not answered

Marked out of 1.00

Where does translation take place in human cells?

Select one:

- Nucleus
- Mitochondria
- Smooth endoplasmic reticulum
- Rough endoplasmic reticulum
- Golgi complex

Your answer is incorrect.

This answer is not precise, but best of the rest. In fact it takes place in ribosomes - Rough ER harbours these ribosomes on its outer surface when translation takes place.

The correct answer is: Rough endoplasmic reticulum

Question 106

Not answered

Marked out of 1.00

Smooth pursuit eye tracking dysfunction is proposed as a strong endophenotype marker for which of the following group of disorders?

Select one:

- Personality disorders
- OCD spectrum disorders
- Mania
- Schizophrenia
- Anxiety disorders

Your answer is incorrect.

Explanation:

The inability to follow a moving visual target accurately is the defining basis for the disorders of smooth visual pursuit and disinhibition of saccadic eye movements seen in patients with schizophrenia. Eye movement dysfunction may be a trait marker for schizophrenia; it is independent of drug treatment and clinical state and also occurs in first-degree relatives of probands with schizophrenia.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 359.

The correct answer is: Schizophrenia

Question 107

Not answered

Marked out of 1.00

Which of the following personality disorders has the strongest support for a genetic contribution?

Select one:

- Dependent
- Schizoid
- Obsessive-compulsive
- Antisocial
- Schizotypal

Your answer is incorrect.

Explanation:

Schizotypal disorder is currently viewed as representing 'partial expression' of the schizophrenia phenotype. Schizophrenia twin studies show an increased risk of schizotypy in the unaffected twin, schizotypy is more common in first-degree relatives of schizophrenic subjects than the general population, and relatives of schizotypal subjects show an increased risk of schizophrenia.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 228-229.

The correct answer is: Schizotypal

Question 108

Not answered

Marked out of 1.00

A child with a deficiency of hypoxanthine-guanine phosphoribosyltransferase developed increased acid. Which of the following behaviours is consistent with this syndrome?

Select one:

- Self injury
- Stereotypes
- Tangential speech
- Visual neglect
- Isolated performance IQ defect

Your answer is incorrect.

Explanation:

Lesch-Nyhan syndrome is an extremely rare recessive condition due to a mutation in the **HGPRT gene** on the **short arm of the X chromosome.** Deficiency of the enzyme hypoxanthine phosphoribosyl transferase leads to **build of uric acid** in the body fluids; partial HGPRT deficiency results in **gout. Early features** include hypotonia and delayed motor milestones; extra pyramidal signs such as spasticity and choreo-athetoid movements develop at about 9 months; hyperreflexia and clonus appear at about 1 year. **Severe compulsive self-mutilation** usually presents between the ages of 2-3; biting of the lips, inside of mouth, fingers.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 816.

The correct answer is: Self injury

Question 109

Not answered

Marked out of 1.00

Mutations that occur in body cells may not get transmitted via gametes. These are called

Select one:

- polymorphisms
- missense mutations
- somatic mutations
- pseudomutations
- nonsense mutations

Your answer is incorrect.

Mutations that occur in somatic cells are somatic mutations and unlike germ cell mutations these may not be transmitted to next generation.

The correct answer is: somatic mutations

Question 110

Not answered

Marked out of 1.00

Trinucleotide repeat expansions in DNA can be detected using

Select one:

- Northern blot
- ELISA
- Polymerase chain reaction
- Western blot
- Southern blot

Your answer is incorrect.

Explanation:

Developed by and named after Edward Southern, a Southern blot is used to identify if a known DNA sequence is present in a sample by using a specific probe to identify the sequence in a large mixture of other DNA fragments. As trinucleotide repeats are DNA abnormalities, the Southern blot can be used to detect them.

Ref: Carter M, Essner R, Goldstein N, Iyer M. Guide to Research Techniques in Neuroscience, 3rd Edition. 2022. p. 239.

The correct answer is: Southern blot

Question 111

Not answered

Marked out of 1.00

ApoE polymorphisms are associated with risk of Alzheimer's disease. Apo E4 alleles are also implicated in following disorders except

Select one:

- Sporadic CJD
- Parkinson disease
- Vascular dementia
- Traumatic brain injury
- Multiple sclerosis

Your answer is incorrect.

Explanation:

There is a clear correlation between the ApoE e4 gene and elevated risk of Alzheimer disease (AD). Although the linkage is not as strong as with AD, ApoE is also associated with progression in other neurological or neurodegenerative diseases, including Parkinson disease, vascular dementia, multiple sclerosis, traumatic brain injury, and ischaemic stroke.

Ref: Yin Y, Wang Z. ApoE and neurodegenerative diseases in aging. Aging and Aging-Related Diseases. 2018:77-92.

The correct answer is: Sporadic CJD

Question 112

Not answered

Marked out of 1.00

Which of the following statements about epidemiology of autism is true?

Select one:

- It is more common in children who had MMR vaccine
- The heritability of this disorder is 80-90%
- The monozygotic twin concordance is more than 95%
- It is seen among children of parents from high socioeconomic status
- It is more common in girls than boys

Your answer is incorrect.

Explanation:

The heritability of autism spectrum disorder is approximately 80-90%.

Ref: Sandin S et al. The heritability of autism spectrum disorder. Jama. 2017 Sep 26;318(12):1182-4.

The correct answer is: The heritability of this disorder is 80-90%

Question 113

Not answered

Marked out of 1.00

Which of the following cells can be described as macrophages of the central nervous system?

Select one:

- Microglia
- Oligodendrocytes
- Amacrine cells
- Endothelium
- Astrocytes

Your answer is incorrect.

Explanation:

There are three types of glial cells. The most common are the astrocytes, which have several functions, including nutrition of neurons, deactivation of some neurotransmitters, and integration with the blood-brain barrier. Mature astrocytes also express some genes enriched in cell progenitors, suggesting they can retain proliferative potential. Theoligodendrocytes in the central nervous system and the Schwann cells in the peripheral nervous system wrap their processed around neuronal axons, resulting in myelin sheaths that facilitate the conduction of electrical signals. The third type of glial cells, the **microglia,** which are derived from **macrophages**, are involved in **removing cellular debris** following neuronal death. The functions of microglia in the non-diseased brain probably include a role in synaptic maintenance.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 924.

Siracusa R et al. Astrocytes: role and functions in brain pathologies. Frontiers in pharmacology. 2019 Sep 27;10:1114.

Sasaki A. Microglia and brain macrophages: an update. Neuropathology. 2017 Oct;37(5):452-64.

The correct answer is: Microglia

Question 114

Not answered

Marked out of 1.00

Which of the following arteries supplies Broca's area?

Select one:

- Posterior cerebral artery
- Anterior cerebral artery
- Vertebral artery
- Middle cerebral artery
- Lenticulostriate artery

Your answer is incorrect.

Explantion:

The most common site of stroke is the middle cerebral artery, an artery that supplies a range of areas involved in both language and motor functions, including Broca’s area.

Ref: Anderlini D et al. Language as a predictor of motor recovery: the case for a more global approach to stroke rehabilitation. Neurorehabilitation and neural repair. 2019 Mar;33(3):167-78.

For a short illustrative description see the video below:

The correct answer is: Middle cerebral artery

Question 115

Not answered

Marked out of 1.00

Which one among the following is not a function of astrocytes?

Select one:

- Nutrition of neurons
- Integration with blood-brain barrier
- Deactivation of neurotransmitters
- Regulation of ion concentration in extra cellular space
- Myelination of axons

Your answer is incorrect.

Explanation:

There are three types of glial cells. The most common are the **astrocytes**, which have several functions, including **nutrition of neurons, deactivation of some neurotransmitters**, and **integration with the blood-brain barrier**. Mature astrocytes also express some genes enriched in cell progenitors, suggesting they can retain proliferative potential. The **oligodendrocytes** in the central nervous system and the **Schwann cells** in the peripheral nervous system wrap their processed around neuronal axons, resulting in **myelin** sheaths that facilitate the conduction of electrical signals. The third type of glial cells, the **microglia,** which are derived from macrophages, are involved in removing cellular debris following neuronal death. The functions of microglia in the non-diseased brain probably include a role in synaptic maintenance.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 924.

Siracusa R et al. Astrocytes: role and functions in brain pathologies. Frontiers in pharmacology. 2019 Sep 27;10:1114.

Sasaki A. Microglia and brain macrophages: an update. Neuropathology. 2017 Oct;37(5):452-64.

The correct answer is: Myelination of axons

Question 116

Not answered

Marked out of 1.00

The part of the brain involved in reward processing is

Select one:

- Amygdala
- Brain stem
- Nucleus accumbens
- Prefrontal cortex
- Hippocampus

Your answer is incorrect.

Explanation:

The midbrain ventral tegmental area (VTA) lies medial to the substantia nigra and contains dopaminergic neurons that give rise to the mesocorticolimbic dopamine system. These neurons send ascending projections that innervate limbic structures, such as the nucleus accumbens and amygdala; the mesoaccumbens pathway is a central element in the neural representation of reward.

The amygdala - an almond-shaped group of nuclei at the heart of the telencephalon - has been associated with a range of cognitive functions. Most prior work on amygdala function emphasized its role in negative emotions, such as fear, and in linking negative emotions with other aspects of cognition, such as learning and memory. However, recent evidence supports a role for the amygdala in processing positive emotions as well as negative ones, including learning about the beneficial biological value of stimuli. Indeed, the amygdala's role in stimulus-reward learning might be just as important as its role in processing negative affect and fear conditioning. In particular, neurons in this region appear to be sensitive to background reward (contingency) and code reward amount and timing, and signal economic decision processes. Primate research also indicates that amygdala may simulate economic decisions of social partners.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 955.

Grabenhorst F, Schultz W. Functions of primate amygdala neurons in economic decisions and social decision simulation. Behavioural Brain Research. 2021 Jul 9;409:113318.

The correct answer is: Nucleus accumbens

Question 117

Not answered

Marked out of 1.00

Which of the following is the only cranial nerve without thalamic relay?

Select one:

- Trigeminal
- Optic
- Abducens
- Oculomotor
- Olfactory

Your answer is incorrect.

The olfactory nerve does not pass through the thalamus, unlike other cranial nerves.

The correct answer is: Olfactory

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